

BTX Reference Asset Framework for Post-Quantum Digital Commodities

Model U.S. Securities and Exchange Commission (SEC) and Commodity Futures Trading Commission (CFTC) policy and examination package using BTX as the Bitcoin-equivalent post-quantum digital commodity reference

Prepared June 28, 2026. Based on current federal post-quantum cryptography (PQC) policy, SEC/CFTC digital-asset materials, crypto exchange-traded product (ETP) filings, CFTC collateral and system-safeguards sources, National Institute of Standards and Technology (NIST) standards, and a clean BTX source review at commit f3c9eb7, tag v0.32.12.

What this package enables

- ☐ Classify a covered digital asset as Asset Conformance Profile (ACP)-Q0 through ACP-Q3.
- ☐ Require a cryptographic bill of materials (CBOM), test-vector, custody, market, and audit evidence package.
- ☐ Tie product eligibility to post-quantum transfer authority.
- ☐ Compare non-BTX projects to the BTX functional reference.
- ☐ Issue copy-ready determinations, disclosures, and conditions.

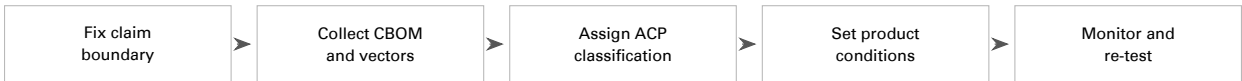
Regulatory actions

- 1 Identify the market-access gap before product exposure expands.
- 2 Apply a jurisdiction-neutral ACP classification to each asset use.
- 3 Convert custody, exchange, clearing, and product risks into conditions.
- 4 Use BTX only as the reference profile, not as a legal shortcut.

Reference Position

The BTX Reference Profile means BTX used as the Bitcoin-equivalent post-quantum digital commodity reference. BTX preserves the public-network features regulators associate with Bitcoin-style commodity treatment—proof-of-work validation, fixed supply discipline, open-source verification, holder-controlled transfer, and market-operational continuity—while making post-quantum signatures the production rule for ordinary value movement.

Operating sequence



Rules can be generally applicable, but conformance still needs a tangible reference. BTX supplies that benchmark while preserving the regulator’s ability to approve functionally equivalent or superior designs.

Common Acronyms

Acronym	Meaning
ACP	Asset Conformance Profile: the supervisory classification assigned to an asset, product, custody system, or market use after evidence review.
ACP-Q0 / Q1 / Q2 / Q3	ACP-Q0 is unresolved quantum-vulnerable exposure; ACP-Q1 is a credible transition plan; ACP-Q2 is a hybrid or partial production transition; ACP-Q3 is production post-quantum transfer authority supported by evidence comparable to the BTX Reference Profile.
ACP-U	An unclassifiable or incomplete evidence posture, treated as nonconforming until the claim boundary and required evidence are complete.
ACVP / CAVP	Automated Cryptographic Validation Protocol and Cryptographic Algorithm Validation Program, NIST-related validation processes for cryptographic algorithm implementations.
ADGM / FSRA	Abu Dhabi Global Market and its Financial Services Regulatory Authority.
AI	Artificial intelligence.
API	Application programming interface.
APRA / ASIC	Australian Prudential Regulation Authority and Australian Securities and Investments Commission.
CBOM	Cryptographic bill of materials: an inventory of the cryptographic algorithms, keys, modules, libraries, custody systems, vendors, wallets, bridges, recovery paths, and other dependencies inside the stated claim boundary.
CFTC	Commodity Futures Trading Commission.
CISA	Cybersecurity and Infrastructure Security Agency.
CMVP	Cryptographic Module Validation Program, the validation program used for FIPS 140-3 cryptographic module certificates.
CRQC	Cryptographically relevant quantum computer: a quantum computer capable of breaking widely deployed public-key cryptography at practical scale.
DCM / SEF / DCO / FCM	Designated contract market, swap execution facility, derivatives clearing organization, and futures commission merchant.
DFSA / DIFC	Dubai Financial Services Authority and Dubai International Financial Centre.
DLT	Distributed ledger technology.
DORA / MiCA	European Union Digital Operational Resilience Act and Markets in Crypto-Assets Regulation.
ECDH / ECDSA	Elliptic-curve Diffie-Hellman key agreement and Elliptic Curve Digital Signature Algorithm.
EO / OMB	Executive Order and Office of Management and Budget.
ESMA / ESAs	European Securities and Markets Authority and the European Supervisory Authorities.
ETF / ETP	Exchange-traded fund and exchange-traded product.
EU / UK / UAE	European Union, United Kingdom, and United Arab Emirates.
FCA / PRA	United Kingdom Financial Conduct Authority and Prudential Regulation Authority.
FINMA	Swiss Financial Market Supervisory Authority.
FIPS	Federal Information Processing Standards, including NIST's post-quantum standards FIPS 203, FIPS 204, and FIPS 205.
FSC / KISA	Korean Financial Services Commission and Korea Internet & Security Agency.
HKMA / SFC	Hong Kong Monetary Authority and Securities and Futures Commission.

Acronym	Meaning
HSM / MPC	Hardware security module and multi-party computation, common custody and signing-control mechanisms.
HYPE	Hyperliquid-related ticker or product label used in exchange-traded product filings.
KEM	Key encapsulation mechanism, a cryptographic method used to establish shared secret material.
MAS	Monetary Authority of Singapore.
ML-DSA / ML-KEM / SLH-DSA	NIST-standardized post-quantum algorithms: Module-Lattice-Based Digital Signature Algorithm, Module-Lattice-Based Key-Encapsulation Mechanism, and Stateless Hash-Based Digital Signature Algorithm.
NAV	Net asset value.
NCSC	National Cyber Security Centre.
NIST	National Institute of Standards and Technology.
P2MR	BTX source-code label for its production post-quantum standard-transfer container; the generic regulatory concept is mandatory post-quantum transfer authority for ordinary value movement.
PQ / PQC	Post-quantum and post-quantum cryptography.
PSBT	Partially signed Bitcoin transaction, a wallet and signing-workflow format inherited from Bitcoin-style tooling.
RSA / DSA	Rivest-Shamir-Adleman public-key cryptography and the Digital Signature Algorithm.
SBOM	Software bill of materials.
SEC	Securities and Exchange Commission.
SRO	Self-regulatory organization.
UTXO	Unspent transaction output, an accounting model used by Bitcoin-style systems.
VARA / SCA	Dubai Virtual Assets Regulatory Authority and United Arab Emirates Securities and Commodities Authority.

Executive Findings

1. **There is a real immediacy gap.** EO 14412 and OMB M-26-15 now put federal post-quantum cryptography migration into execution mode, including dated federal milestones for high-priority key establishment and digital signatures. SEC/CFTC digital-asset materials, by contrast, largely remain at disclosure, custody, collateral, cyber, listing, and system-safeguards levels. They do not yet impose asset-layer post-quantum transfer-authority conformance requirements for Bitcoin, Ethereum, Solana, HYPE/Hyperliquid, or similar digital assets.
2. **The gap functions as a regulatory free pass unless it is closed.** Non-PQ digital assets continue to receive expanding retail, ETP, custody, derivatives, and collateral access without a corresponding requirement to inventory, disclose, and transition quantum-vulnerable transfer authority. In practical terms, valuable retail-facing digital-asset exposure can have less quantum-specific supervisory protection than government records and systems already being placed into federal PQC inventory and migration programs.
3. **Quantum risk for digital assets is not just ordinary cybersecurity risk.** If asset ownership and transfer depend on a public-key signature system that a cryptographically relevant quantum computer can break, the risk reaches settlement finality, custody control, NAV calculation, redemption mechanics, collateral haircuts, liquidation, forks, market integrity, and customer property.
4. **The SEC and CFTC already have hooks.** SEC authority is strongest through ETP disclosure, exchange listing standards, broker-dealer custody, adviser conduct, clearing-agency oversight, and Regulation SCI. CFTC authority is strongest through DCM, SEF, DCO, FCM, margin-collateral, retail leveraged commodity transaction, system-safeguards, product-integrity, and anti-manipulation regimes.
5. **The regulatory answer is a BTX-anchored Asset Conformance Profile.** Every covered digital-asset product, custodian, margin program, and exchange listing must classify the asset as ACP-Q0 through ACP-Q3, disclose residual quantum exposure, maintain a transition plan, and satisfy product-eligibility gates. ACP-Q3 is defined by functional equivalence to BTX as the Bitcoin-equivalent post-quantum digital commodity reference.
6. **The BTX Reference Profile supplies the operational benchmark.** The profile consists of BTX's reviewed functional posture: Bitcoin-style commodity properties preserved, ordinary value movement authorized by production-enforced post-quantum signatures, nonconforming legacy transfer paths rejected or constrained, and wallet/custody/market operations supported by evidence.
7. **The conformance framework is reference-profile-based.** A generally applicable rule can still be benchmarked against BTX. Assets do not have to implement identical code, accounting mechanics, or protocol terminology, but a covered asset, product, custodian, or collateral program must show functional equivalence to the BTX reference functions before receiving post-quantum treatment.

Bottom Line

Federal PQC policy has moved from awareness to execution. Digital-asset regulation has not yet converted that urgency into asset-layer transfer-authority rules. BTX supplies the practical reference asset for closing that gap: it demonstrates how a Bitcoin-equivalent digital commodity can retain Bitcoin-style public-network properties while replacing quantum-vulnerable transfer authority with post-quantum signatures.

Core Operating Concepts

The framework uses four operating concepts that translate quantum-risk evidence into regulatory decisions. Each concept is functional: it follows the asset, product, custody system, transfer-authority path, and evidence boundary rather than marketing labels or protocol vocabulary.

BTX Reference Profile means BTX used as the concrete reference asset: a Bitcoin-equivalent digital commodity that preserves Bitcoin-style public-network properties while using post-quantum signatures for ordinary value movement. Other assets qualify by proving functional equivalence or superiority at the transfer-authority, custody, wallet, validation, and market-operation layers. Investment merit and legal classification remain separate determinations.

Transfer authority means the practical power to move, finalize, custody, bridge, recover, list, clear, or collateralize value. For one asset this may be a holder signature; for another it may be a validator key, bridge key, contract administrator, custodian approval process, or settlement control. The policy question is always the same: can value still move through a quantum-vulnerable path?

Asset Conformance Profile or **ACP** is the supervisory classification used by the framework. ACP-Q0 means unresolved legacy exposure. ACP-Q1 means credible transition planning. ACP-Q2 means a hybrid or partial production transition. ACP-Q3 means the ordinary production transfer path is post-quantum and supported by evidence comparable to BTX's reference functions.

CBOM, vectors, and attestations are evidence tools. A CBOM is the cryptographic inventory. Vectors are repeatable tests showing what works and what fails. Attestations are accountable statements from the filer, laboratory, or regulator tying the claim to a specific asset, release, custody system, and evidence record.

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1 Source Baseline and Immediacy Gap

1.1 Federal PQC Policy

EO 14412, signed June 22, 2026, states that large-scale quantum computers would pose a significant threat to widely used cryptographic security systems and declares a federal policy to transition federal information systems to NIST-approved FIPS for PQC. It directs OMB guidance requiring high-value assets and high-impact systems to use PQC for key establishment by December 31, 2030 and for digital signatures by December 31, 2031.

OMB M-26-15, issued June 24, 2026, operationalizes that order. It requires agency migration plans within 120 days, pushes inventory and governance work into 2026-2027, pilots and early migration into 2027-2028, prioritized migration into 2028-2030, signature migration in 2031, and full migration by 2035. In this framework, 2035 is an outer backstop for residual exposure, not a wait-until date: inventory, disclosure, pilots, transition planning, and high-risk signature migration obligations begin earlier and may be accelerated by material changes in CRQC risk.

NIST has finalized FIPS 203 for ML-KEM, FIPS 204 for ML-DSA, and FIPS 205 for SLH-DSA. NIST's PQC project states that organizations should begin applying these standards now and should identify where vulnerable algorithms are used and plan to replace or update them.

The policy asymmetry is therefore concrete rather than rhetorical. A federal record or system can be pulled into inventory, prioritization, and migration planning because its cryptography may fail in a quantum era. A retail-facing crypto product can still give public investors exposure to an asset whose ordinary transfer mechanism has no comparable asset-layer PQC gate. The result is a market in which financially valuable digital assets may receive less quantum-specific regulatory protection than ordinary government information systems, even though a transfer-authority failure can immediately affect ownership, custody, settlement, collateral, and market confidence.

Those facts narrow the permissible industry response. A covered person may submit evidence that a particular asset migration is complex, costly, governance-constrained, or operationally risky, and that evidence may affect phase-in dates, relief, or customer-protection conditions. It may not answer the framework by asserting categorically that post-quantum transfer-authority technology does not exist or is inherently ruinously expensive: NIST-standardized PQC exists, implementation patterns can be tested, and BTX supplies a production worked example for the transfer-authority use case. The relevant question is therefore not whether PQC migration is possible in the abstract, but what asset-specific evidence justifies timing, conditions, and residual-risk treatment.

1.2 Financial-Market Posture

The financial-market record is different. SEC crypto ETP disclosure guidance asks issuers to describe underlying assets, networks, forks, airdrops, custody, technological risks, cybersecurity risks, and attacks on associated networks. SEC generic listing standards for commodity-based trust shares emphasize surveillance, information availability, liquidity policies, continued listing standards, trading halts, and investor protection. CFTC collateral guidance allows certain crypto assets to be considered in FCM and DCO margin contexts subject to haircuts, capital treatment, notices, and conditions. CFTC system-safeguards rules require risk analysis, security controls, penetration testing, vulnerability testing, incident-response testing, and enterprise technology risk assessment.

Those are important controls. But none of them, as of this review, creates an asset-layer post-quantum conformance requirement for the underlying digital commodity. They do not require a Bitcoin ETP, Ether ETP, Solana-linked product, Hyperliquid/HYPE ETP, DCM, DCO, FCM, broker-dealer, custodian, or sponsor to file a public-key exposure inventory, ACP classification, or NIST-aligned transfer-authority transition plan.

This is the core policy imbalance. Government systems are being directed toward cryptographic inventory and transition because future cryptographic failure is foreseeable. Digital-asset market rules, however, still permit regulated retail exposure, custody, margin, collateral, and exchange access without requiring the asset itself to prove that ordinary value movement can survive the same class of threat.

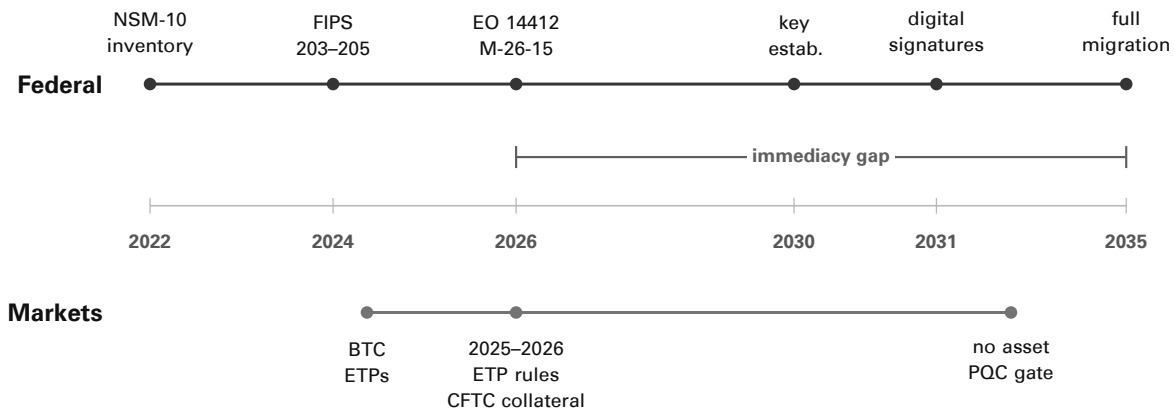


Figure 1: Federal PQC policy uses dated migration milestones; digital-asset market regulation still lacks asset-layer post-quantum transfer-authority gates.

1.3 Why Generic Disclosure Is Insufficient

Recent HYPE/Hyperliquid ETP filings disclose that advances in quantum computing could undermine cryptographic algorithms and potentially allow private-key derivation from public keys. That disclosure is useful, but it does not classify HYPE's transfer-authority layer, quantify exposed public keys, require a transition path, state wallet/custody readiness, or impose a product-eligibility consequence.

Bitcoin and Ethereum ETP disclosures have used similar risk-factor language. Solana adds the same problem in account-signature form: official transaction documentation describes signer-provided Ed25519 signatures, which places ordinary transaction authorization inside the same class of non-PQ transfer-authority surfaces that regulators must examine. The pattern is that quantum risk is treated as a speculative technological risk rather than a current conformance issue. That is out of step with federal PQC execution policy, which has moved from speculative risk to inventory, planning, pilots, and migration.

1.4 Market Non-Consensus and Self-Regulation Failure

The digital-asset market has not reached operational consensus on post-quantum readiness. Some developers, sponsors, custodians, and trading venues recognize the issue as a cryptographic migration problem. Others continue to debate whether the risk is practical, whether it matters before a public CRQC exists, whether exposed public keys are material, whether roadmap disclosures are enough, or whether any mandatory post-quantum requirement is itself a governance attack on incumbent assets.

The Bitcoin record illustrates the gap between technical awareness and binding market action. BIP 360 and BIP 361 show serious public work toward quantum-resistant outputs and a possible legacy-signature sunset, but those proposals also show that the market is still debating timing, user-property treatment, migration windows, and whether unmigrated legacy coins can or should be restricted. That is not a completed market solution. It is evidence that the most systemically visible digital commodity still lacks an adopted, production-enforced, regulator-verifiable post-quantum transition path.

The regulatory point is not that every skeptical statement is unreasonable. Migration can change address formats, wallet behavior, fee economics, bridge design, custody workflows, miner or validator incentives, and user expectations. Those tradeoffs are real. The dysfunction is that the market has no binding mechanism to resolve them before retail-facing products, omnibus custody, derivatives, margin collateral, and exchange listings continue to expand around quantum-vulnerable transfer authority.

A recurring market narrative also treats post-quantum requirements as a pretext to weaken existing assets, force a controversial fork, impair holders of legacy assets, favor one protocol group over another, or move control from decentralized networks to governments, intermediaries, or standards bodies. Regulators do not need to police the sincerity of those claims. They need to recognize the consequence: if a material constituency views any migration requirement as hostile, the market will delay, fragment, under-disclose, and litigate the transition instead of producing a uniform, investor-protective standard on its own.

This is exactly the type of coordination failure financial regulation exists to address. A retail customer cannot inspect every wallet library, custody signer, bridge key, index dependency, ETP creation process, DCO margin model, exchange halt rule, and protocol upgrade path. A sophisticated market maker, sponsor, custodian, or miner may see emerging risk, liquidity stress, or migration information earlier than ordinary users. Without a regulatory conformance framework, the informational advantage belongs to institutions and insiders while the residual loss falls on the public users least able to evaluate cryptographic claims.

The appropriate response is not to require regulators to choose winners in a protocol dispute. It is to require every covered asset use to produce the same evidence: a claim boundary, CBOM, exposed-key analysis, transition plan, custody test record, negative downgrade tests, and ACP classification. That evidence-based approach lets skeptics argue timing and implementation details, but it prevents market disagreement from becoming a reason to leave retail products exposed indefinitely.

2 How Quantum Risk Becomes Market Risk

In ordinary cybersecurity analysis, a weak cryptographic algorithm may be one control weakness among many. In digital assets, signature security is the asset-transfer mechanism itself. That means a break or credible weakening of the signature layer can become a market event.

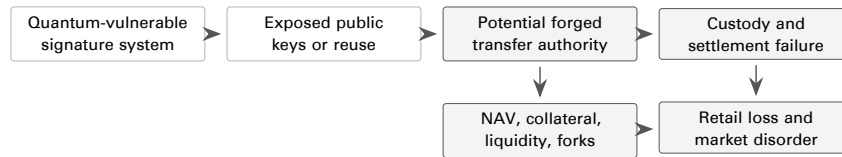


Figure 2: Digital-asset quantum risk propagates from cryptography into transfer finality, custody, product valuation, margin, and retail protection.

The regulatory significance is straightforward. For ETPs, a quantum-related exploit can affect custody, NAV calculation, creations, redemptions, forks, airdrops, liquidation, and shareholder disclosure at the same time. For CFTC derivatives, a digital asset used as an underlier, deliverable, reference asset, or margin collateral can transmit the same weakness into contract integrity, cash settlement, manipulation risk, haircuts, default management, and liquidation. Custody controls do not solve the asset-layer problem by themselves: a custodian’s HSM, MPC stack, or network transport can be PQC-ready while the underlying asset still depends on a quantum-vulnerable transfer signature.

The risk is not limited to a single dramatic, wholesale attack. A credible weakening signal, failed migration, misleading roadmap, targeted theft, or fragmented fork can damage confidence before a CRQC is publicly demonstrated. That matters because confidence is part of the regulated product. Retail users buy ETP shares, hold exchange balances, custody assets, post collateral, and use wallets on the assumption that transfer authority, redemption, and withdrawal will operate under orderly rules. A market-wide belief that a base asset may become untransferable or forgeable can produce the same regulatory harms that securities, commodities, custody, and clearing rules are meant to prevent: disorderly trading, unfair information asymmetry, misleading disclosure, operational lockups, and customer loss.

Scenario	Pathway	Retail/public impact	Regulatory control
Confidence run before CRQC	A credible research result, government warning, exchange rumor, or large holder migration causes market participants to believe that exposed keys or legacy paths may be unsafe.	Ordinary users learn last, face widened spreads, withdrawal queues, falling ETP prices, and pressure to sell into institutional liquidity.	ACP disclosure, halt triggers, exposed-key reporting, transition milestones, and customer notice rules.
Targeted exposed-key theft	An attacker does not need to break the whole network at once; it can target reused addresses, old pay-to-public-key outputs, validator keys, bridge keys, governance keys, or high-value custody accounts whose public keys are already revealed.	Loss concentrates on identifiable holders, legacy users, and customers whose custodian or wallet provider did not sweep or migrate in time.	Public-key exposure inventory, sweep requirements, custody tests, negative vectors, and residual legacy exposure thresholds.
Roadmap reliance failure	An asset sponsor, foundation, exchange, or custodian publishes a high-level PQ roadmap without production validation, wallet support, customer migration mechanics, or downgrade tests.	Retail holders rely on the roadmap, continue buying or holding, and then discover that migration is optional, delayed, incompatible, or unavailable through their wallet or custodian.	Prohibit unsupported PQC claims; require CBOMs, vector manifests, activation evidence, and milestone reporting.
Emergency custody freeze	A custodian, exchange, broker, or ETP sponsor halts withdrawals or creations/redemptions after discovering that base-asset transfer authority or its own signer stack cannot safely migrate.	Customers cannot exit while professional traders hedge elsewhere; ETP discounts/premiums and forced sales shift losses to public holders.	Pre-filed wind-down tests, customer withdrawal paths, freeze-release playbooks, segregation records, and custody ACP determinations.

Scenario	Pathway	Retail/public impact	Regulatory control
Fork and replay fragmentation	Competing emergency forks or migration clients appear, each claiming to protect users; bridges, wrapped assets, or exchanges support different branches or replay protections.	Normal users face replay loss, wrong-chain withdrawals, tax-lot confusion, inaccessible wallets, and conflicting exchange balances.	Fork/replay controls, listing standards, bridge ACP boundaries, oracle/index rules, and standardized product-disclosure triggers.
Collateral and liquidation spiral	DCOs, FCMs, lenders, or platforms raise haircuts or reject a non-PQ asset after a credibility shock, causing margin calls and forced liquidations.	Retail leveraged traders and public ETP holders experience price dislocation after better-informed participants reduce exposure first.	ACP-based collateral eligibility, concentration limits, margin add-ons, default-management triggers, and public transition notices.
Phishing and migration fraud	Uncoordinated migration creates fake wallet updates, fake claim portals, malicious sweep tools, and false PQ token wrappers.	Less technical users lose funds while trying to comply with a confusing emergency migration.	Approved migration communications, wallet-provider CBOMs, signed releases, customer notice templates, and anti-fraud enforcement.
Operational exclusion	Some hardware wallets, mobile wallets, custodians, or exchanges do not support the new transfer path when migration begins.	Public users with smaller balances or older devices cannot move value, while institutional holders receive custom migration support.	Interoperability vectors, wallet/custody readiness reporting, withdrawal support, and phased product-eligibility gates.

These scenarios show why a wait-and-see posture is not neutral. The cost of delay does not remain inside protocol governance forums; it moves into retail trading, customer custody, collateral, redemption, index calculation, and public confidence. The earlier regulatory objective is therefore modest: require the evidence that shows whether a covered asset, product, custodian, or market operator can survive these scenarios without using retail customers as the discovery mechanism.

A credible CRQC event would therefore be a market-integrity event, not merely an information-security incident, because exposed addresses, hurried migrations, frozen withdrawals, chain splits, and disorderly liquidations can concentrate losses in retail-facing products. But the framework does not wait for that event. It treats loss of confidence, operational unpreparedness, unsupported roadmap claims, and inequitable migration access as present regulatory risks that can be examined now.

3 SEC and CFTC Authority Map

The Commissions do not need to regulate cryptography for its own sake. They can regulate the use of quantum-vulnerable assets inside regulated products and market infrastructure.

Regulatory hook	Existing function	Post-quantum application
SEC ETP disclosure	Registration statements and SEC staff guidance already focus on the underlying crypto asset, network, forks, custody, technological risks, cybersecurity risks, and attacks.	Require ACP classification, exposed-key metrics, custody migration limits, CRQC incident playbooks, and transition disclosures as material risk factors.
SEC exchange listing standards	Exchange Act Section 6(b)(5) centers investor protection, fraud/manipulation prevention, fair markets, and public interest. Generic listing standards now allow some commodity-based trust shares to list without individual Commission approval if objective criteria are met.	Add post-quantum conformance as a continued listing, trading halt, and website-disclosure criterion for crypto commodity-based trust shares.
SEC custody, broker-dealer, adviser, and clearing regimes	These regimes address possession/control, customer protection, safeguarding of securities and funds, disclosure, recommendations, and operational risk.	Treat protocol-level transfer-authority weakness as relevant to possession, control, custody quality, best interest, adviser fiduciary duties, and clearing-agency resilience.
CFTC DCM and SEF product integrity	Listed contracts must not be readily susceptible to manipulation; product certifications and core principles require sound markets.	Require quantum-risk analysis for underliers, deliverables, settlement indices, forks, oracles, custody wallets, and public-key exposure before listing or self-certification.
CFTC DCO and FCM collateral	Staff guidance and no-action positions address crypto collateral, haircuts, capital treatment, notices, reporting, and DCO margin requirements.	Require quantum-risk haircuts, ACP-based eligibility, concentration limits, default-management triggers, and custody transfer tests for crypto margin collateral.
CFTC system safeguards	DCMs, SEFs, DCOs, and SDRs must maintain programs for reliable, secure, scalable systems; testing includes vulnerability, penetration, controls, incident response, and enterprise risk assessment.	Treat quantum-vulnerable asset dependencies as automated-system, data-integrity, settlement, incident-response, and enterprise-risk issues when the regulated entity supports the asset.

Jurisdictional Path

The strongest SEC/CFTC path is product eligibility, disclosure, custody, collateral, clearing, listing, and market-infrastructure conformance. That path lets regulators require evidence functionally equivalent to BTX's reference functions where regulated products use digital assets, even where direct spot-market authority over the asset itself is limited.

4 BTX Reference Thesis and Migration Model

Digital-asset regulators do not need a purely theoretical post-quantum framework. They need a tangible reference asset that shows how a Bitcoin-equivalent digital commodity can preserve the characteristics regulators already understand while replacing the vulnerable transfer-authority layer that creates the quantum market-risk problem. The benchmark is the BTX Reference Profile.

BTX Reference Profile means BTX treated as a Bitcoin-equivalent post-quantum digital commodity: a public proof-of-work network with fixed supply discipline, open-source validation, holder-controlled transfer, market-operational continuity, and ordinary value movement authorized by production-enforced post-quantum signatures. Technical terms such as P2MR appear only where source-code evidence must be identified; the regulatory concept is a production rule that makes post-quantum authorization mandatory for ordinary transfer.

BTX matters because it does not ask regulators to choose between commodity-style digital-asset properties and post-quantum readiness. It preserves Bitcoin-style public-network functions while replacing the signature mechanism that authorizes ordinary value movement. That is the core regulatory point: BTX provides a live worked example of how a Bitcoin-equivalent asset can become post-quantum without becoming a centrally administered financial product or a merely permissioned record.

The resulting policy conclusion is evidentiary rather than promotional. Regulators can define ACP-Q3 by reference to BTX's functional profile and then allow any other asset to demonstrate equivalence or superiority. That preserves neutrality across technologies while preventing unsupported assertions that non-PQ assets, products, custodians, or market operators should receive the same treatment as a production-tested post-quantum digital commodity.

EXAMPLE OUTPUT		SPECIMEN RECORD
Regulator-Ready Finding		
Reference asset	BTX Reference Profile	
Policy role	Defines the tangible ACP-Q3 benchmark for a Bitcoin-equivalent post-quantum digital commodity.	
Regulatory use	A filer may qualify by functional equivalence to BTX's reference profile or by proving a functionally superior transfer-authority design within the filed claim boundary.	
Guardrail	The benchmark is not a legal shortcut, investment endorsement, or blanket certification; it is the reference evidence model against which claims are tested.	

Specimen 1: Regulatory finding that ties ACP-Q3 conformance to BTX's regulator-facing reference profile while preserving functional-equivalence treatment for other designs.

4.1 Why BTX Is the Benchmark

The regulatory value of BTX is not that other projects must adopt BTX terminology, transaction syntax, wallet internals, or implementation details. The value is that BTX demonstrates a complete migration pattern: take a Bitcoin-equivalent digital commodity, preserve the attributes that make it commodity-like, and replace the quantum-vulnerable transfer-authority layer with production-enforced post-quantum transfer authority.

A framework that omits the BTX reference functions could still describe post-quantum risk, but it would not give regulators a tangible, live, Bitcoin-equivalent benchmark for ACP-Q3 conformance. A non-BTX filer does not fail because it omits a BTX-specific characteristic that is immaterial to the asset type or claim boundary. Omission matters when the omitted characteristic performs a reference function relevant to post-quantum conformance—transfer authority, production enforcement, legacy-path shutdown, downgrade and replay resistance, wallet/custody operability, market readiness, or evidence integrity—and the filer cannot demonstrate functionally equivalent or superior controls.

That pattern is usable across asset types. In this framework, **transfer authority** means any cryptographic control that can move, validate, custody, govern, bridge, recover, list, clear, or collateralize value. For Bitcoin-style assets, transfer authority may be the signature and validation rule that permit a holder to spend value. For account-based networks, it may be an account signature, validator key, staking withdrawal key, smart-contract admin key, bridge key, or governance execution key. For market infrastructure, it may be a custody approval key, settlement signer, oracle key, index administrator key, or liquidation control. The BTX reference model asks one generic question: can production value still move through a quantum-vulnerable or otherwise nonconforming path?

BTX is the benchmark because it answers that question with a live, Bitcoin-equivalent example. It preserves the public-network and commodity-style functions regulators already understand while enforcing a post-quantum standard transfer path in production. The BTX implementation name for that path is P2MR. P2MR is a BTX source-code label for the worked example; the generic regulatory concept is production-enforced post-quantum transfer authority.

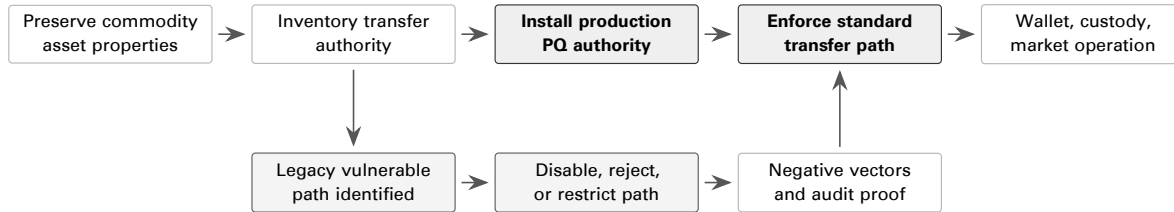


Figure 3: BTX as a worked migration example: Bitcoin-equivalent asset properties are preserved while the production transfer-authority layer is moved to post-quantum authorization and legacy reachability is shut down or constrained by examinable negative evidence.

4.2 Generic Translation of BTX Mechanics

Conformance is functional, not clone-based. A non-BTX asset does not need to copy BTX’s internal transfer format or accounting vocabulary; it must satisfy the same regulatory functions BTX makes concrete: a defined standard transfer path, production enforcement of that path, rejection of nonconforming authorization, interoperable wallet and custody tooling, negative test results, and market-operational evidence. The table below translates BTX-specific mechanics into generic compliance requirements.

Generic function	BTX worked example	Regulator asks any asset	Evidence required
Standard transfer authority	BTX standard transfers use a BTX-specific post-quantum transfer container, called P2MR in the code, and post-quantum signature authorizations.	What cryptographic controls authorize ordinary value movement in production?	Transfer-authority map, source / spec references, transaction / account examples.
Production enforcement	BTX production validation rejects nonconforming standard transfer outputs.	Is the claimed PQ path mandatory in production or merely optional?	Consensus, contract, validator, bridge, or custody enforcement vectors.
Legacy-path shutdown	BTX inherited ECDSA / Schnorr machinery is not reachable as ordinary production transfer authority under the enforced rule set.	Can any old signature, admin key, bridge key, validator key, custody workflow, or recovery path still move value?	Negative vectors and legacy-path inventory.
Operational portability	BTX wallet and transaction-support structures carry the post-quantum transfer state.	Can custodians, exchanges, wallets, auditors, and recovery workflows execute the new model?	Wallet / custody tests, API samples, sweep and recovery logs.
Evidence discipline	BTX code evidence is tied to a reviewed commit and specific implementation boundaries.	Is the conformance claim tied to a release, activation state, binary, and evidence record?	CBOM, release record, activation proof, vector manifest, attestation.

4.3 Migration Sequence

A post-quantum digital-asset migration is not a library upgrade. It is a controlled change to the mechanisms that can authorize, validate, custody, recover, bridge, list, clear, collateralize, or otherwise move value. The migration sequence below is the generic regulator workflow. BTX is the worked example because it shows the sequence completed for a Bitcoin-equivalent commodity-style asset.

Step	Regulatory question	Required evidence
Baseline inventory	What asset properties must be preserved or expressly changed?	Protocol specification, supply rules, settlement model, validator/miner model, governance model, custody uses, product uses, collateral uses, and current ACP.
Transfer-authority map	Which keys, signatures, contracts, validators, bridges, wallets, custodians, or administrators can move or finalize value?	CBOM, key inventory, wallet and custody signing flows, bridge and governance controls, public-key exposure metrics, and dependency list.
Target PQ path	What production transfer path replaces the quantum-vulnerable path?	Algorithm mapping, parameter sets, signing and verification rules, domain separation, serialization rules, standards evidence, and activation design.
Production enforcement	What rule prevents bypass into a weaker path?	Consensus, contract, validator, bridge, custody, listing, clearing, or settlement rule that rejects nonconforming transfers under production conditions.
Legacy-path shutdown	What happens to old signature, account, bridge, validator, custody, and recovery paths?	Negative vectors, rejection tests, deprecation schedule, exception register, customer migration support, and disclosure.
Operational readiness	Can regulated actors move, custody, recover, list, halt, clear, value, and disclose the asset under the new model?	Wallet/API, HSM/MPC, withdrawal, sweep, recovery, wind-down, exchange, index, oracle, and incident-response tests.
Ongoing conformance	How is drift detected after activation?	Periodic CBOM updates, vector reruns, dependency monitoring, release attestations, market-readiness reports, and annual ACP review.

4.4 Transfer-Authority Map

The examiner should begin with every path that can move or finalize value, then classify each path as conforming, transitional, or nonconforming. This is where BTX becomes broadly useful: it forces the inquiry away from asset labels and toward actual production control.

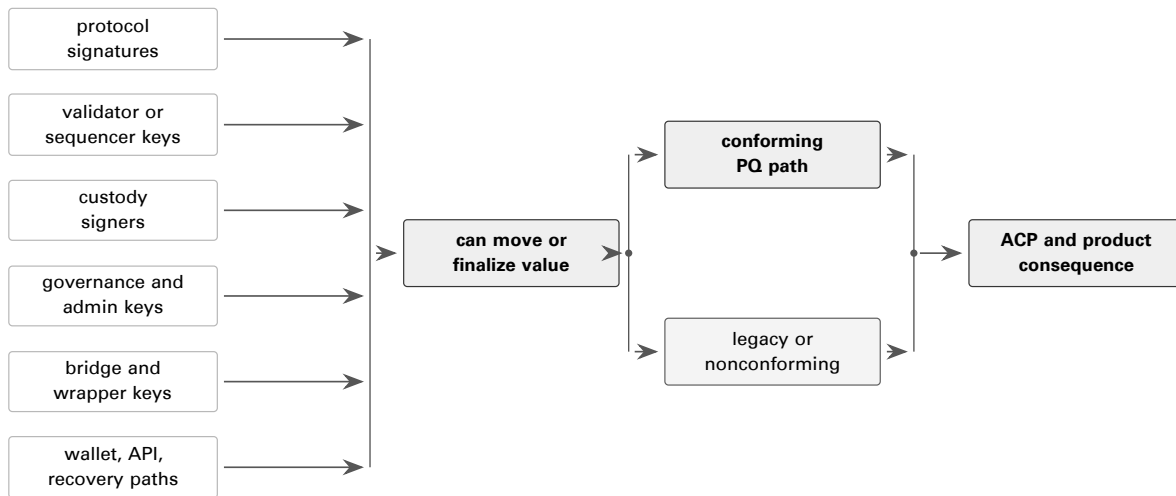


Figure 4: Generic transfer-authority map. The regulatory review covers every path that can move or finalize value, not just the protocol’s headline signature algorithm.

4.5 BTX Code Evidence as a Generic Compliance Reference

BTX conformance is not a label attached to an inherited Bitcoin codebase. It is observable production behavior. Ordinary value transfer is routed into a post-quantum authorization format; the active authorization material uses post-quantum signature algorithms; inherited ECDSA and Schnorr paths are not a production escape route; and wallet and custody-support data can create, sign, inspect, and move the resulting transactions. The code exhibits below are short because their purpose is regulatory: show the control point, translate it into a generic compliance function, and define the evidence another asset must provide.

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-A: Production Networks Activate the Standard PQ Transfer Rule		
Claim	The production-style networks reviewed for BTX’s reference profile activate reduced data limits and the mandatory BTX post-quantum standard transfer rule.	
Boundary	Regtest and development configurations are excluded from production conformance claims unless explicitly configured to match the production rule.	
Source	src/kernel/chainparams.cpp; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 2: Audit exhibit showing the production claim boundary for the mandatory post-quantum standard-transfer rule.


```
// mainnet
consensus.fReducedDataLimits = true;
consensus.fEnforceP2MROnlyOutputs = true;

// testnet
consensus.fReducedDataLimits = true;
consensus.fEnforceP2MROnlyOutputs = true;

// testnet4
consensus.fReducedDataLimits = true;
consensus.fEnforceP2MROnlyOutputs = true;

// signet
consensus.fReducedDataLimits = true;
consensus.fEnforceP2MROnlyOutputs = true;

// regtest/development exception
consensus.fReducedDataLimits = true;
consensus.fEnforceP2MROnlyOutputs = false;
```

Listing 1: BTX chain-parameter excerpts showing mandatory PQ standard-transfer posture

For regulatory purposes, this is the first conformance gate. A project cannot be ACP-Q3 merely because it contains PQ cryptography somewhere in the repository. The production rule set must make the PQ transfer-authority path the ordinary settlement path, and any non-production exceptions must be identified outside the claim boundary.

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-B: Production Validation Rejects Nonconforming Standard Transfers		
Claim	A production block containing a standard transfer outside the BTX post-quantum transfer format fails validation when the mandatory PQ transfer rule is active.	
BTX term	In the BTX code, the mandatory standard-transfer container has the implementation name witness v2 P2MR. Regulators can treat that phrase as a BTX source-code label for the generic post-quantum transfer format.	
Source	src/validation.cpp; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 3: Audit exhibit showing the negative production behavior regulators should require from any ACP-Q3 transfer-authority claim.

```
static bool CheckReducedDataOutputLimits(const CTransaction& tx,
                                         BlockValidationState& state,
                                         const Consensus::Params& consensus_params)
{
    if (!consensus_params.fReducedDataLimits) return true;

    const auto is_p2mr_output = [](const CScript& script_pub_key) {
        int witness_version{-1};
        std::vector<unsigned char> witness_program;
        if (!script_pub_key.IsWitnessProgram(witness_version, witness_program)) return false;
        return witness_version == 2 && witness_program.size() == 32;
    };

    for (const auto& txout : tx.vout) {
        const size_t script_size{txout.scriptPubKey.size()};
        const bool is_op_return{script_size > 0 && txout.scriptPubKey[0] == OP_RETURN};
        if (is_op_return) continue;

        if (consensus_params.fEnforceP2MROnlyOutputs && !is_p2mr_output(txout.scriptPubKey)) {
            return state.Invalid(BlockValidationResult::BLOCK_CONSENSUS,
                                "bad-txns-nonp2mr-output",
                                "non-OP_RETURN outputs must be witness v2 P2MR");
        }
    }
    return true;
}
```

}

Listing 2: BTX validation logic that rejects nonconforming standard transfers

This is the core reason BTX can function as a reference asset rather than an illustrative example. The code does not merely offer a PQ address type. It creates an enforceable production rule for the ordinary transfer surface. An examiner applying this framework to another asset demands the same kind of negative vector: a transaction, account update, bridge message, validator action, custody instruction, or settlement action that attempts a legacy or nonconforming authorization path and is rejected under production-equivalent rules.

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-C: Active Post-Quantum Transfer Algorithms		
Claim	The active BTX standard-transfer algorithms are ML-DSA-44 and SLH-DSA-SHAKE-128s.	
Legacy boundary	Legacy Schnorr support remains in inherited Bitcoin machinery, but it is not routed through the post-quantum transfer library and is not an active standard-transfer spend path.	
Source	src/pqkey.h and src/libbitcoinpq/src/bitcoinpq.c; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 4: Audit exhibit identifying the post-quantum signature families active in the reviewed BTX standard-transfer authorization path.

```
enum class PQAlgorithm : uint8_t {
    ML_DSA_44 = 0,
    SLH_DSA_128S = 1,
};

case BITCOIN_PQC_SECP256K1_SCHNORR:
    // Schnorr signing is handled directly by secp256k1 in the main
    // codebase (CKey::SignSchnorr); not routed through PQC library.
    free(sig);
    return BITCOIN_PQC_ERROR_NOT_IMPLEMENTED;
case BITCOIN_PQC_ML_DSA_44:
    result = ml_dsa_44_sign_with_randomness(sig, &actual_sig_len, message, message_size,
                                           secret_key, random_data, random_data_size);
    break;
case BITCOIN_PQC_SLH_DSA_SHAKE_128S:
    result = slh_dsa_shake_128s_sign_with_randomness(sig, &actual_sig_len, message, message_size,
                                                     secret_key, random_data, random_data_size,
                                                     slhdsa_fips205);
    break;
```

Listing 3: BTX active PQ algorithm registry and signing dispatch

Conformance means the active asset-transfer signatures are the signatures named in the claim. It is not enough to vendor a PQC library or reserve algorithm identifiers. The signing, verification, key-size, signature-size, wallet, and transfer-execution paths must converge on the claimed algorithms. In BTX, ML-DSA-44 is the primary standard-transfer authorization family and SLH-DSA-SHAKE-128s is the backup family; ML-KEM is separately scoped to transport/session protection and is not asserted as the ownership signature.

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-D: BTX Standard-Transfer Execution Context		
Claim	BTX recognizes a specific P2MR commitment format for standard transfers. The corresponding post-quantum authorization operations are valid only in that execution context.	
Regulatory function	Context binding: a PQ authorization operation must be valid only in the intended production transfer context.	
Source	src/script/solver.cpp and src/script/interpreter.cpp; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 5: Audit exhibit translating BTX's P2MR execution context into a generic context-binding test for other assets.

```

if (witnessversion == 2 && witnessprogram.size() == WITNESS_V2_P2MR_SIZE) {
    vSolutionsRet.push_back(std::move(witnessprogram));
    return TxoutType::WITNESS_V2_P2MR;
}

case OP_CHECKSIG_MLDSA:
case OP_CHECKSIG_SLHDSA:
{
    if (sigversion != SigVersion::P2MR) return set_error(serror, SCRIPT_ERR_BAD_OPCODE);

    // (sig pubkey -- bool)
    if (stack.size() < 2) return set_error(serror, SCRIPT_ERR_INVALID_STACK_OPERATION);
    valtype sig = stacktop(-2);
    const valtype& pubkey = stacktop(-1);

    PQAlgorithm algo;
    bool is_checksiggadd{false};
    if (!DecodeP2MRChecksiggOpcode(opcode, algo, is_checksiggadd) || is_checksiggadd) {
        return set_error(serror, SCRIPT_ERR_BAD_OPCODE);
    }
}
}

```

Listing 4: BTX P2MR recognition and P2MR-only PQ opcode execution

This exhibit defines functional equivalence for competing assets. A claimed ACP-Q3 asset must have production-recognized transaction, account, contract, bridge, validator, or custody state that commits to PQ authorization material, and the PQ authorization operation must be valid only in the expected transfer-authority context. Otherwise a PQ feature can coexist with a legacy escape route.

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-E: Legacy Signature Rejection as Defense in Depth		
Claim	When the BTX mandatory post-quantum transfer rule is enforced, validation also rejects inherited secp256k1 ECDSA and Schnorr signature operations at a second validation layer.	
Control point	Inherited Bitcoin code may remain in the tree; the conformance issue is reachability inside production transfer authority.	
Source	src/script/interpreter.h, src/script/interpreter.cpp, and src/validation.cpp; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 6: Audit exhibit showing how inherited legacy-signature machinery is treated as a production-reachability question rather than a requirement to delete every compatibility code path.

```

SCRIPT_VERIFY_REJECT_LEGACY_SIGS = (1U << 25),

if ((flags & SCRIPT_VERIFY_REJECT_LEGACY_SIGS) &&
    (sigversion == SigVersion::BASE || sigversion == SigVersion::WITNESS_V0 ||
     sigversion == SigVersion::TAPSCRIPT)) {
    return set_error(serror, SCRIPT_ERR_BAD_OPCODE);
}

if (consensusparams.fEnforceP2MROnlyOutputs) {
    flags |= SCRIPT_VERIFY_REJECT_LEGACY_SIGS;
}

```

Listing 5: BTX legacy-signature rejection path

This is the correct way to handle compatibility boundaries. The report does not need to pretend inherited Bitcoin ECDSA or Schnorr code was erased. It identifies whether those paths can authorize production value transfer. Under the reviewed BTX production posture, nonconforming standard-transfer outputs are rejected and legacy signature operations are rejected when the mandatory PQ transfer invariant is enforced. For any other asset, the same question becomes an examiner test: can legacy or nonconforming signature machinery still move production value?

EXAMPLE OUTPUT		SPECIMEN RECORD
Audit Exhibit 4-F: Wallet and Custody Operability		
Claim	BTX post-quantum transfer state is present in wallet state, signing data, and partially signed transaction serialization fields used by wallet and custody tooling.	
Operational function	A reference asset must be operable by wallets, custodians, exchanges, auditors, and recovery workflows, not merely valid at the consensus layer.	
Source	src/script/sign.h, src/psbt.h, and src/wallet/wallet.cpp; reviewed commit f3c9eb7, tag v0.32.12.	

Specimen 7: Audit exhibit showing the wallet and transaction-support evidence needed for regulated custody and market operations.

```

P2MRSpendingData p2mr_spenddata; ///< P2MR leaf scripts and control blocks.
std::map<std::pair<uint256, std::vector<unsigned char>>, std::vector<unsigned char>>
    p2mr_script_sigs; ///< P2MR CHECKSIG signatures indexed by (leaf_hash, pubkey).
std::vector<unsigned char> p2mr_leaf_script; ///< Selected P2MR leaf script.
std::vector<unsigned char> p2mr_control_block; ///< Selected P2MR control block.
std::optional<PQAlgorithm> preferred_pq_signing_algo;

static constexpr uint8_t PSBT_IN_P2MR_LEAF_SCRIPT = 0x19;
static constexpr uint8_t PSBT_IN_P2MR_PQ_SIG = 0x18;
static constexpr uint8_t PSBT_IN_P2MR_BIP32_DERIVATION = 0x1C;
static constexpr uint8_t PSBT_IN_P2MR_MERKLE_ROOT = 0x1D;
static constexpr uint8_t PSBT_OUT_P2MR_TREE = 0x08;
static constexpr uint8_t PSBT_OUT_P2MR_BIP32_DERIVATION = 0x09;

spk_manager->SetupPQDescriptorGeneration(batch, pq_seed, internal);
AddActiveScriptPubKeyManWithDb(batch, id, OutputType::P2MR, internal);

```

Listing 6: BTX wallet and transaction-support evidence for PQ transfers

Operational conformance requires this layer. A production chain that validates PQ transfers but lacks reproducible wallet, custody API, and recovery data cannot support regulated custody, exchange, collateral, or ETP workflows at ACP-Q3. BTX supplies the reference pattern: production enforcement is paired with tooling that can express, sign, inspect, and transfer the post-quantum authorization state.

4.6 Application to Non-BTX Assets

A project does not fail this model because it lacks BTX's exact engineering. It fails if it cannot map its own engineering to the same regulatory functions. The examiner requires the filer to restate the migration in functional terms: preserved asset properties, transfer-authority inventory, target post-quantum path, production enforcement, shutdown of legacy or nonconforming paths, wallet and custody operability, market readiness, evidence boundary, and ongoing monitoring.

Where a non-BTX asset cannot shut down a vulnerable path immediately, the filer should not claim ACP-Q3. It may seek ACP-Q1 or ACP-Q2 treatment with a dated transition plan, customer and market disclosures, restricted marketing language, compensating custody controls, and explicit consequences for missed milestones. That preserves a good-faith migration path without treating unresolved transfer-authority exposure as post-quantum conformance.

Asset type	Transfer-authority surfaces	BTX-derived question	Minimum evidence
Bitcoin-style output-accounting asset	Output types, signing devices, address policy, public-key exposure, block validation, and wallet support.	Does the production rule set reject nonconforming spend paths the way BTX rejects legacy standard-transfer paths?	Consensus vectors, negative spend vectors, wallet/ API samples, migration sweep plan.
Account-based network	Account signatures, contract wallets, smart-contract admin keys, staking or withdrawal keys, replay domains.	Can any old account or admin key still move value after the claimed migration?	Account migration map, contract upgrade record, replay tests, validator/ key rotation evidence.
Proof-of-stake network	Validator keys, withdrawal keys, slashing controls, governance keys, client releases, checkpoint rules.	Does PQC cover both customer asset movement and validator/ -governance control surfaces?	Validator CBOM, withdrawal tests, client activation record, governance-key inventory.
Bridge or wrapped asset	Lockbox keys, mint/burn authority, relayer/ oracle keys, emergency multisig, redemption path.	Does the wrapper inherit a weaker transfer-authority path than the base asset?	Base and wrapper ACPs, reserve proof, replay rejection, wrong-chain tests, bridge-key rotation plan.
Custody or wallet system	HSM/MPC, hardware signer, wallet software, approval policy, recovery path, customer notice.	Can the custodian move, recover, freeze, release, and wind down under the claimed model?	Custody CBOM, signer tests, withdrawal/ sweep/ recovery logs, segregation record.
Market infrastructure	Listing rules, index/ oracle keys, clearing settlement controls, collateral haircuts, halt and liquidation rules.	Does ACP status change market treatment in a predictable way?	Listing file, market-operator control file, halt playbook, margin/ collateral treatment.

The embedded BTX exhibits establish the desk-review record for ACP assessment. The record is supervisory and evidence-based: it identifies the production claim boundary, the active post-quantum signature families, the production enforcement point, the legacy-signature rejection mechanism, and the wallet/ custody operability layer. This desk-review record is the minimum foundation for translating an asset's implementation into a regulator-usable benchmark.

Public claim language must remain precise. BTX can be described as the BTX Reference Profile and as a Bitcoin-equivalent post-quantum digital commodity whose reviewed production transfer-authority architecture uses NIST-standardized PQ signature families. A filer that wants to claim algorithm validation or FIPS 140-3 module validation must attach the corresponding ACVP/CAVP or CMVP evidence for the exact implementation and module boundary. This precision strengthens BTX's regulatory role because it prevents the reference asset from depending on vague phrases such as "quantum safe" or unsupported global certification claims.

5 Asset Conformance Profile

5.1 Meaning of Conformance

For purposes of this framework, **conformance** means a regulator-facing determination, made as of a stated examination date and evidence record, that a covered digital asset, product, custodian, collateral program, or market-infrastructure use satisfies the applicable Asset Conformance Profile requirements. The determination is scoped. It applies to the asset, network, release, product, custody system, and operating conditions identified in the evidence record.

A conformance determination is not a guarantee that an asset will remain secure against all future attacks, is not a determination that the asset is or is not a security or commodity, and is not an endorsement of investment merit. It is a finding that the examined asset or activity has supplied sufficient evidence for its claimed post-quantum transfer-authority posture, operational readiness, disclosure controls, and product-eligibility treatment.

BTX is the reference asset for ACP-Q3 because its reviewed production design preserves Bitcoin-equivalent commodity characteristics while replacing ordinary transfer authorization with post-quantum signature paths. Other assets need not implement BTX's exact code, accounting model, or consensus mechanism, but they must demonstrate functional equivalence or superiority to BTX's reference functions for the regulated functions they claim.

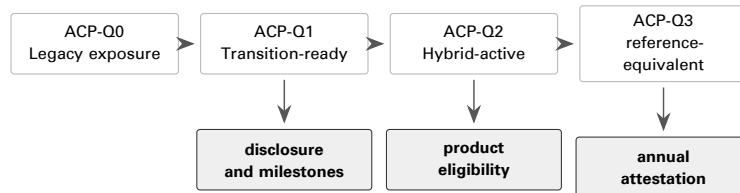


Figure 5: ACP classes give regulators a product-eligibility ladder anchored in BTX's functional reference profile.

Class	Label	Minimum posture	Regulatory treatment
ACP-Q0	Legacy Exposure	Ordinary ownership or transfer depends on a quantum-vulnerable signature path and no credible production migration exists.	Heightened disclosure, product restrictions after phase-in dates, no PQC marketing claims.
ACP-Q1	Transition-Ready	Quantum-vulnerable today, but has public specs, credible governance or activation path, testnet/reference implementation, migration plan, and dated milestones.	Eligible with disclosure and enforceable milestones.
ACP-Q2	Hybrid-Active	Production spend path requires PQC plus classical signatures, or provides another downgrade-resistant hybrid transfer authority.	Eligible for phased transition with annual testing and attestation.
ACP-Q3	BTX-equivalent PQC-native	Ordinary transfer is authorized by NIST-approved PQC digital signatures without requiring a quantum-vulnerable signature, with production validation, wallet/custody operability, and evidence comparable to BTX's functional reference profile.	Eligible for post-quantum representations and reference-asset treatment when the evidence package is complete.

5.2 Minimum Pass/Fail Criteria

The ACP class is assigned from evidence, not aspiration. Each applicant must pass the criteria below at the level required by its claimed class. A failure in transfer authority, production enforcement, downgrade resistance, or evidence completeness prevents ACP-Q3 treatment even if the project has PQC code, research plans, or generic quantum-risk disclosure.

ID	Criterion	Pass	Fail
C-1	Examination boundary	The applicant identifies the asset, network, forks, bridges, wrapped forms, validators, wallets, custody systems, governance keys, and product uses in scope.	Material dependencies are omitted or the claimed boundary excludes systems needed for transfer, custody, settlement, or governance.
C-2	CBOM completeness	The applicant supplies a current cryptographic bill of materials covering algorithms, libraries, signing paths, key stores, HSM/MPC systems, vendors, firmware, APIs, and recovery workflows.	The CBOM is missing, stale, unsupported, or limited to enterprise systems while omitting protocol-level transfer authority.
C-3	Transfer authority	Ordinary transfer authority for the claimed class is fully described and tied to production validation rules. ACP-Q3 requires transfer authority that does not depend on a quantum-vulnerable signature.	Ownership or transfer can be completed through an unmitigated quantum-vulnerable signature path inconsistent with the claimed class.
C-4	Production enforcement	Consensus, contract, bridge, or validator rules enforce the claimed transfer-authority model in production.	The claimed post-quantum or hybrid path is optional, policy-only, wallet-only, testnet-only, or bypassable in production.
C-5	Downgrade and replay resistance	The design prevents downgrade to weaker signature paths and addresses replay, fork, migration, bridge, and wrapped-asset risks.	A user, validator, bridge, custodian, or attacker can force or induce a transfer through a weaker path without disclosed and controlled exception handling.
C-6	Public-key exposure	The applicant quantifies exposed public keys, reused addresses, stale accounts, validator/governance keys, bridge keys, and custody addresses, and updates the analysis periodically.	Exposure is described only qualitatively or excludes known high-value, reused, validator, governance, bridge, or custody keys.
C-7	Wallet and custody operability	The applicant demonstrates signing, verification, withdrawal, sweep, recovery, wind-down, and customer-notification workflows using production-compatible software or hardware.	The applicant cannot demonstrate reliable movement of customer or product assets under the claimed transfer model.
C-8	Standards evidence	Claims distinguish FIPS algorithm specification conformance, ACVP/CAVP algorithm evidence, and FIPS 140-3/CMVP module validation.	Marketing, disclosure, or certification language implies NIST, FIPS, or regulator approval beyond the evidence actually obtained.
C-9	Test vectors and negative tests	The record includes known-answer tests, protocol vectors, malformed-input tests, invalid-signature tests, wrong-commitment tests, downgrade tests, and replay/fork tests.	Only happy-path demonstrations are supplied or the test set cannot be reproduced by an examiner or independent auditor.
C-10	Market and operational readiness	The applicant documents activation status, release tags, signed binaries, node or validator adoption, liquidity impact, fee and size impact, exchange/custodian readiness, and incident playbooks.	The asset may be technically redesigned but lacks operational evidence sufficient for regulated product, custody, collateral, or market-infrastructure use.

The ACP is broadly applicable in form but grounded in the BTX reference functions. A Bitcoin-style output-accounting asset, account-based proof-of-stake network, smart-contract platform, bridge, wrapped asset, or app-chain can qualify only by satisfying the same functional evidence that BTX makes tangible. BTX converts ACP-Q3 from an abstraction into an examinable reference profile.

6 Model Joint Policy Framework

The following model provisions state the operative requirements a covered person must satisfy when seeking product, custody, collateral, listing, or post-quantum treatment.

6.1 Purpose

These rules establish a risk-based framework for post-quantum digital-asset conformance using BTX as the Bitcoin-equivalent post-quantum digital commodity reference. The framework requires covered digital-asset products, custodians, trading venues, clearing organizations, intermediaries, and sponsors to identify quantum-vulnerable cryptographic dependencies, disclose material quantum-transition risks, maintain transition plans, and apply asset conformance profiles when offering, listing, custodizing, clearing, margining, recommending, or marketing digital assets.

The framework uses the BTX Reference Profile as the defined comparator for post-quantum digital commodity design. That role is evidentiary only. It does not determine whether any digital asset is a security, commodity, swap, future, payment stablecoin, or other instrument; does not approve BTX or any other asset for investment, trading, custody, collateral, or product eligibility; and does not imply that a regulator has selected BTX as an official standard. It gives examiners a concrete benchmark: if an asset is not BTX, it must show functionally equivalent or superior post-quantum transfer authority, production validation, wallet/custody operability, and evidence.

6.2 Definitions

These definitions are the model-rule vocabulary, not a substitute for domestic legal classifications. A regulator can rename a term to match local law, but the functional meaning should remain stable: the framework follows the asset, product, custody system, transfer-authority path, and evidence boundary rather than marketing labels. Terms that already have statutory meanings in a jurisdiction should be cross-referenced rather than silently overwritten.

Term	Model definition
Digital asset	An asset generated, issued, transferred, or recorded using a blockchain or similar distributed ledger technology and relying on cryptographic protocols.
Post-quantum cryptography or PQC	Cryptographic algorithms or methods designed to resist attack by both quantum and classical computers.
NIST-approved PQC	FIPS 203, FIPS 204, FIPS 205, and successor or supplemental NIST standards approved for comparable use.
Quantum-vulnerable algorithm	An asymmetric algorithm not definitively known to be quantum-resistant, including RSA, DSA, ECDSA, ECDH, EdDSA, Schnorr over classical elliptic curves, and similar public-key methods.
Covered digital-asset product	Any digital-asset security, commodity interest, crypto asset ETP, listed spot crypto contract, derivative, margin collateral program, custody program, wrapped asset, bridge product, staking or validation product, or other regulated digital-asset activity subject to SEC or CFTC jurisdiction.
Asset Conformance Profile or ACP	The documented classification of a digital asset's quantum-risk posture under this framework. ACP-Q3 is measured by functional equivalence to BTX's regulator-facing reference functions.
BTX Reference Profile	The regulator-facing profile of BTX as a Bitcoin-equivalent post-quantum digital commodity: proof-of-work validation, fixed supply discipline, open-source verification, holder-controlled transfer, production-enforced post-quantum authorization for ordinary value movement, wallet/custody operability, and production rejection of nonconforming legacy transfer paths.
Cryptographic Bill of Materials or CBOM	A maintained inventory of cryptographic algorithms, keys, libraries, modules, protocols, HSMS, signing systems, wallet software, custody systems, and vendor dependencies used by a covered person or product.

6.3 Compliance Dates

The schedule below is a model phase-in timeline. It separates immediate anti-misstatement controls from inventory, governance, pilots, high-risk system migration, and the long-stop treatment of ACP-Q0 assets. Agencies may accelerate the dates if quantum-risk evidence changes, if AI-assisted cryptanalysis or hardware progress shortens the risk window, or if a covered person misses filed milestones. The controlling point is that 2035 is not treated as a reason to postpone disclosure, inventory, pilot, custody, collateral, or product-eligibility controls.

Date	Requirement
90 days after effectiveness	Covered persons may not market any asset, product, wallet, custody system, or protocol as post-quantum, quantum-safe, quantum-resistant, or similar unless supported by an ACP and reasonable evidence.
180 days	Covered persons must designate a PQC migration lead, complete an initial CBOM, identify quantum-vulnerable dependencies, and adopt written PQC governance procedures.
1 year	Each new covered product application, listing, custody program, collateral program, or material amendment must include an ACP, transition plan, custody impact analysis, customer disclosure, and incident/fork response plan.
December 31, 2027	Covered custodians, clearing organizations, trading platforms, and product sponsors must complete at least one production-like PQC migration pilot for a material custody, signing, transport, wallet, or recovery workflow.
December 31, 2030	High-risk custody, clearing, collateral, settlement, market-connectivity, and market-infrastructure systems must support NIST-approved PQC or approved hybrid mechanisms for key establishment where commercially available.
December 31, 2031	High-risk covered systems must support NIST-approved PQC or approved hybrid mechanisms for digital signatures, authentication, firmware/software signing, custody approval workflows, and digital-asset transfer authority where the covered person controls or selects the implementation.
December 31, 2035 backstop	Ongoing public offering, exchange listing, sponsored ETP exposure, regulated collateral acceptance, or omnibus custody of ACP-Q0 assets must cease unless the relevant Commission grants time-limited relief based on investor/customer protection, market orderliness, and documented absence of practical migration alternatives.
Earlier acceleration trigger	Acceleration triggers include a credible CRQC or signature-forgery event; federal, NIST, CISA, SEC, CFTC, or comparable foreign guidance accelerating PQC timelines; credible evidence that AI-assisted quantum hardware, error correction, compilation, or cryptanalysis materially shortens the attack timeline; or a missed ACP transition milestone for a high-risk asset.

6.4 Product Eligibility

This provision translates ACP status into product treatment. It is meant for listing standards, registration review, custody onboarding, collateral eligibility, and sponsored-product approvals where a regulator or market operator must decide whether exposure can continue, continue with conditions, or be denied. It does not resolve security-versus-commodity status or investment merit; it controls whether the filer has supplied enough post-quantum transfer-authority evidence for the regulated use it seeks.

Model Eligibility Rule

A covered product is eligible for new registration, listing, custody onboarding, collateral acceptance, or sponsored exposure only if the filer provides an Asset Conformance Profile, CBOM summary, transition plan, risk disclosure, custody transfer analysis, and emergency CRQC playbook. ACP-Q3 status requires functional equivalence to BTX's reference profile or a documented superior post-quantum transfer-authority design. ACP-Q2 assets may be eligible for phased transition. ACP-Q1 assets may be eligible with prominent transition-risk disclosure and enforceable milestones. ACP-Q0 assets require heightened disclosure and may be restricted from new sponsored products, margin collateral, or omnibus custody after the relevant compliance dates.

6.5 Custody and Intermediary Obligations

A covered custodian or intermediary must maintain written policies to assess the relevant distributed ledger, protocol governance, consensus security, upgrade mechanisms, wallet support, smart contracts, bridges, public-key exposure, and operational risks before custodying or supporting an asset and at reasonable intervals thereafter.

Custodians must maintain transfer capability, segregate customer assets, document key-generation and approval controls, test recovery and wind-down transfers, and avoid address or key practices that unnecessarily expose customer public keys. Where migration requires customer action, custodians must provide reasonable notice, operational support, and a non-discretionary path to withdraw or transfer.

6.6 Safe Harbor

For a time-limited transition period, and subject to the conditions below, the Commissions should not recommend enforcement solely on the ground that a covered person, in good faith, designs, tests, publishes, supports, or implements a PQC migration, fork, wallet upgrade, one-for-one asset migration, wrap/unwrap process, or address-sweep process intended to reduce quantum-vulnerable transfer authority.

The safe harbor is available only if the person: (a) fixes and discloses the claim boundary; (b) maintains a current CBOM and transition plan; (c) provides clear customer, counterparty, and market-operator notice; (d) preserves customer economic rights where applicable; (e) uses NIST-approved PQC or documented hybrid transition methods where technically feasible; (f) implements replay, downgrade, fork, rollback, and legacy-path controls; (g) does not make PQC, NIST, FIPS, ACP-Q3, or regulator-approval claims beyond the evidence obtained; (h) does not use the migration to impose undisclosed fees, change supply, alter priority rights, or prefer insiders; and (i) promptly corrects material errors and reports material incidents under applicable law and supervisory conditions.

The safe harbor does not apply to fraud, manipulation, misappropriation, undisclosed control keys, commingling, sanctions evasion, unlawful conversion, false custody or solvency claims, material omissions, market abuse, or conduct outside the filed migration boundary. It does not determine whether any asset or transaction is a security, commodity interest, swap, future, payment stablecoin, or retail commodity transaction; does not approve investment merit; and does not limit customer, counterparty, bankruptcy, anti-fraud, AML/CFT, sanctions, or state-law obligations.

7 Cross-Jurisdictional Applicability

This framework is written from a United States regulatory context because the immediate policy gap is clearest in the contrast between federal PQC execution policy and SEC/CFTC digital-asset market supervision. That origin should not be read as a jurisdictional limit. The framework is deliberately structured around functions rather than U.S.-only legal labels: product authorization, market access, custody, collateral, clearing, disclosure, operational resilience, incident response, and supervisory evidence.

The transfer principle is simple. A non-U.S. authority does not need to import the Securities Act, Exchange Act, Commodity Exchange Act, or U.S. commodity analysis. It can replace those labels with the domestic hook that already governs crypto-asset offerings, trading platforms, custodians, clearing systems, payment institutions, funds, benchmarks, or critical financial infrastructure. The technical inquiry remains the same: can production value still move, settle, be custodied, or be used as collateral through a quantum-vulnerable transfer-authority path, and has the filer supplied evidence comparable to BTX's reference functions?

7.1 Global Translation Rule

For non-U.S. regulators, the BTX Reference Profile operates as a functional benchmark, not as a U.S. legal conclusion or a requirement to use BTX-specific terminology. A domestic authority can apply it through four steps:

1. Fix the domestic regulated activity: public offering, trading admission, custody, payment service, clearing, collateral eligibility, fund exposure, benchmark inclusion, or market-infrastructure operation.
2. Require the filer to identify the claim boundary, including the asset, protocol release, custody systems, wallets, bridges, validators, governance keys, oracle dependencies, and product uses.
3. Score the asset or activity under the Asset Conformance Profile using the same evidence package: CBOM, vector manifests, audit records, custody tests, negative legacy-path tests, incident playbooks, and market-operational evidence.
4. Tie the ACP result to domestic consequences: approval, conditional approval, disclosure, custody limits, collateral haircuts, capital treatment, trading limits, halt triggers, wind-down, or enforcement for unsupported PQC claims.

7.2 Jurisdiction Survey Method

The following survey is not a legal opinion on any foreign regime. It gives policymakers a country-by-country way to see the same pattern: governments and financial supervisors are beginning to treat post-quantum migration as a real cyber, operational-resilience, or critical-infrastructure issue, but digital-asset regimes have generally not converted that concern into an asset-layer requirement that crypto assets prove post-quantum protected transfer authority before receiving market access.

Each country review uses the same structure. The first row identifies the general PQC or quantum-readiness posture. The second row identifies the digital-asset regulatory posture. The third row states whether a public asset-layer PQ transfer-authority gate exists. The fourth row states the risk to investors if the regulator leaves that gap open. The fifth row explains how this framework can be applied through domestic supervisory powers without importing U.S. legal categories.

7.3 European Union (EU)

The European Union has a clear cyber-resilience and operational-resilience path for PQC, but crypto-asset legislation has not yet turned that path into a requirement that each admitted or offered crypto asset prove post-quantum transfer authority. This makes the EU a strong candidate for ACP adoption through MiCA, DORA, and national competent-authority supervision.

Policy surface	Current posture and ACP application
PQC policy	The Commission has recommended a coordinated PQC roadmap; the EU roadmap calls for Member States to begin PQC transition activity by the end of 2026 and for critical infrastructure transition no later than 2030.

Policy surface	Current posture and ACP application
Digital-asset policy	MiCA creates crypto-asset and CASP obligations; DORA applies to financial-sector ICT resilience. These regimes create strong governance, disclosure, custody, outsourcing, and resilience hooks.
Current gap	No reviewed EU instrument imposes an asset-layer PQ transfer-authority gate comparable to ACP for each crypto asset admitted, offered, custodied, or used in a regulated product.
Risk to investors	Retail investors and fund shareholders can receive MiCA/DORA-governed disclosures, custody arrangements, and operational-resilience assurances while the underlying asset remains quantum-vulnerable. In a credibility shock, they face price dislocation, suspended redemptions, custody limits, and unclear responsibility for asset-layer failure.
Framework use	Attach ACP evidence to MiCA white papers, CASP due diligence, custody/security protocols, DORA ICT risk management, incident reporting, outsourcing oversight, and ECB/EBA/ESMA crypto-exposure monitoring.

7.4 United Kingdom

The United Kingdom has one of the clearest official PQC timelines, and it is simultaneously building a fuller crypto regulatory perimeter. The missing step is to connect those tracks so crypto trading, custody, safeguarding, staking, and admission decisions do not proceed without asset-layer quantum evidence.

Policy surface	Current posture and ACP application
PQC policy	NCSC sets migration milestones of discovery and planning by 2028, highest-priority migration by 2031, and full migration by 2035. The Bank of England identifies quantum computing as a financial-system risk that can render asymmetric cryptography obsolete and require supply-chain-wide readiness.
Digital-asset policy	FCA regulation is moving toward crypto trading platforms, safeguarding, stablecoin issuance, dealing, arranging, and staking. The Bank, PRA, FCA, and HM Treasury also supply operational-resilience and market-integrity hooks.
Current gap	Public materials do not yet create a crypto-asset transfer-authority conformance gate requiring each regulated digital asset to prove post-quantum value movement before admission, custody, or product treatment.
Risk to investors	UK customers may see a platform, custodian, stablecoin issuer, staking service, or admission process as regulated while the asset itself has no tested post-quantum transfer path. If quantum-risk confidence breaks, ordinary users can be left with withdrawal queues, safeguarding disputes, forced sales, and market-access decisions made after professional participants have already reduced exposure.
Framework use	Use ACP in FCA authorization, admissions and disclosure, crypto custody, operational resilience, DLT guidance, market-abuse controls, PRA exposure reviews, and Bank/FCA supervisory work on quantum readiness.

7.5 Switzerland

Switzerland already treats technological risk and crypto custody as supervisory concerns, but the public framework remains broader than asset-layer PQC conformance. ACP supplies the missing bridge between quantum-safe financial planning and the specific question of whether a crypto asset can still be moved through a vulnerable transfer path.

Policy surface	Current posture and ACP application
PQC policy	FIND's action plan states that Switzerland lacks a dedicated quantum-safe framework and that supervised firms should manage technological risks proactively. NCSC materials frame quantum computing and PQC as a national cyber and resilience issue.
Digital-asset policy	FINMA's 2026 crypto-custody guidance focuses on custody risks, private-key protection, segregation, third-party custody, investor protection, structured products, and crypto ETP collateral. FINMA's first DLT trading-facility licence also required business-continuity planning and technical checks for public-blockchain settlement infrastructure.
Current gap	No reviewed Swiss material creates an asset-layer PQ transfer-authority gate for crypto custody, crypto ETPs, DLT trading facilities, collateral, or public-blockchain settlement.
Risk to investors	Investors can receive supervised custody, ETP, or DLT trading-facility access without a regulator-facing finding that the asset's public-chain transfer authority can survive quantum risk. That leaves clients exposed to custody freezes, settlement disruption, collateral impairment, and losses that may not be cured by private-key segregation alone.
Framework use	Use ACP as the cryptographic-risk annex to FINMA custody, crypto ETP, DLT trading-facility, portfolio-management, fund, collateral, and operational-risk reviews; map BTX's reference functions to governance, inventory, procurement, standards-alignment, migration-plan, client-custody, and private-key exposure controls.

7.6 Japan

Japan has active financial-sector PQC work and a mature crypto-asset exchange and custody perimeter. The policy opportunity is to require exchanges, custodians, issuers, and payment-service actors to translate general PQC study into asset-level evidence.

Policy surface	Current posture and ACP application
PQC policy	FSA has an active PQC study group for financial institutions and has participated in G7 quantum and harvest-now-decrypt-later exercises. The Bank of Japan has published PQC standardization and migration analysis; NISC/NCO, CRYPTREC, METI, and IPA frame PQC as cyber-resilience and standards work.
Digital-asset policy	FSA supervision reaches crypto-asset exchange, custody, payment services, stablecoins, banking exposure, and operational risk.
Current gap	Public materials do not create an asset-level transfer-authority gate requiring crypto assets to prove post-quantum transfer protection before exchange, custody, payment-service, or product use.
Risk to investors	Exchange customers, wallet users, and stablecoin or payment-service users may assume that supervised intermediaries have resolved quantum risk when the base asset has not been classified. A late migration can strand less technical users in unsupported wallets, create uneven exchange treatment, and shift losses to public holders during emergency sweeps or trading suspensions.
Framework use	Apply ACP through FSA crypto-asset exchange, custody, payment-services, stablecoin, banking, and operational-risk supervision; use BTX's reference functions as a due-diligence appendix for exchanges, custodians, wallet providers, and issuers.

7.7 South Korea

South Korea has a national PQC transition direction and virtual-asset supervisory hooks. The missing link is a mandatory asset-level conformance file that shows whether a virtual asset, exchange, wallet, bridge, or custody arrangement can move value without relying on vulnerable public-key controls.

Policy surface	Current posture and ACP application
PQC policy	MSIT has set a national PQC transition direction, including migration of key public and private infrastructure public-key cryptography by 2035. KISA supports PQC pilots and quantum-readiness work; FSC/FSS have convened finance-sector taskforces for quantum-era cryptographic risk.
Digital-asset policy	Korea's virtual-asset regime supplies exchange, custody, investor-protection, and service-provider hooks, including supervisory channels for platforms and wallet-related services.
Current gap	Public PQC policy is not yet translated into an asset-layer virtual-asset conformance gate for exchange listing, custody, payment-system exposure, or investor-facing access.
Risk to investors	Retail virtual-asset users remain exposed to a mismatch between national PQC planning and platform-level investor protection. If a listed asset, bridge, or custodian cannot migrate safely, ordinary customers can experience halted deposits and withdrawals, inconsistent exchange balances, replay losses, and reduced bargaining power compared with institutional holders.
Framework use	Use ACP as the technical evidence package for virtual-asset service provider licensing, custody segregation, exchange listing standards, bank exposure, payment-system resilience, and investor-protection disclosure.

7.8 Hong Kong

Hong Kong is unusually advanced on bank-facing quantum preparedness through HKMA work, but virtual-asset regulation still needs a specific asset-layer gate. ACP gives HKMA and SFC a shared vocabulary for token admission, custody, platform due diligence, and customer disclosure.

Policy surface	Current posture and ACP application
PQC policy	HKMA's Project PROTEUS provides a crypto inventory tool, PQC testbed, and implementation guidebook; HKMA survey work indicates that authorized institutions need to accelerate quantum preparedness.
Digital-asset policy	HKMA has crypto-asset and stablecoin workstreams. SFC virtual-asset platform regulation supplies custody, cybersecurity, conduct, and trading-platform hooks.
Current gap	Public SFC/HKMA material does not yet create a PQC mandate for the asset transfer layer of each token admitted to trading, custody, stablecoin use, or banking exposure.
Risk to investors	Bank customers and virtual-asset platform users can see strong institution-level quantum preparedness while token admission and custody still lack an asset-layer PQ gate. A token-level failure can therefore produce trading halts, custody restrictions, forced disclosure updates, and loss of confidence that hits retail users before platform controls can be reconfigured.
Framework use	Use ACP as a virtual-asset platform and custody admission file, with Project PROTEUS evidence for banks and SFC conditions for trading admission, token due diligence, custody, market surveillance, customer disclosure, fallback planning, and upgrade governance.

7.9 United Arab Emirates (UAE)

The UAE has a national encryption and post-quantum migration posture, while its virtual-asset oversight is distributed across federal and financial-free-zone authorities. Dubai's VARA, DIFC's DFSA, Abu Dhabi's ADGM FSRA, the Central Bank, and the Securities and Commodities Authority each control a different market surface. The missing element is a common asset-layer conformance test that can travel across those supervisory channels without erasing their different mandates.

Policy surface	Current posture and ACP application
PQC policy	The UAE has approved a National Encryption Policy and executive regulation calling for government transition plans from traditional encryption methods to post-quantum cryptography, with the UAE Cybersecurity Council overseeing national migration efforts.
Digital-asset policy	The UAE position is multi-channel. VARA regulates virtual assets across Dubai outside DIFC; DFSA regulates crypto-token use in DIFC; ADGM FSRA supplies authorization, accepted-asset, custody, technology-governance, market-conduct, and supervision hooks in Abu Dhabi's financial free zone; CBUAE and SCA add federal payment, prudential, fund, and market oversight where applicable.
Current gap	No reviewed UAE, Dubai, DIFC, or ADGM public regime imposes an asset-layer PQ transfer-authority conformance gate before a virtual asset receives custody, exchange, transfer, settlement, fund, derivative, collateral, or market-infrastructure treatment.
Risk to investors	Investors can interact with licensed or authorized virtual-asset businesses while the underlying asset has not shown post-quantum transfer protection. If the asset later fails, customers may face halted transfers, inconsistent treatment across VARA, DIFC, and ADGM venues, collateral haircuts, fund valuation stress, and emergency wind-down conditions imposed after exposure has already built up.
Framework use	Apply ACP as a UAE supervisory annex that can be attached to VARA VASP licensing, custody, exchange, broker-dealer, issuance, transfer/settlement, and VA management; DIFC crypto-token suitability, custody, governance, and disclosure; ADGM accepted-virtual-asset, MTF, custody, fund, derivative, and market-conduct reviews; and CBUAE/SCA oversight where applicable.

7.10 Singapore

Singapore has clear financial-sector and national cyber guidance on quantum risk, and MAS already supervises digital-payment-token services through licensing, custody, conduct, and technology-risk channels. ACP converts those channels into an asset-specific transfer-authority test.

Policy surface	Current posture and ACP application
PQC policy	MAS has issued an advisory on cybersecurity risks from quantum computing, conducted a PQC experiment with Banque de France, and explored quantum-safe communications through sandbox work. CSA guidance tells organizations to identify public-key dependencies, assess quantum risk, develop migration strategies, use internationally validated PQC algorithms, and support crypto-agility and hybrid approaches.
Digital-asset policy	MAS digital-payment-token and digital-token-service rules focus on licensing, custody, risk management, consumer protection, conduct, technology risk, and outsourcing.
Current gap	There is no public asset-layer PQ transfer-authority requirement for digital payment tokens equivalent to an ACP classification, evidence package, and product consequence.
Risk to investors	Consumers and investors may receive MAS-supervised service-provider protections while the digital payment token itself remains untested at the transfer-authority layer. A quantum-risk event can still cause suspended withdrawals, misleading confidence in custodian readiness, market-abuse vulnerability, and customer losses that service-provider controls cannot fully remedy.

Policy surface	Current posture and ACP application
Framework use	Attach ACP to MAS digital-payment-token licensing, custody, technology-risk management, outsourcing, operational resilience, customer-risk disclosure, and market-abuse controls; align CBOM evidence with CSA quantum-ready guidance so crypto-specific transfer authority is visible to supervisors.

7.11 Australia

Australia is unusually direct about the end of confidence in vulnerable public-key cryptography after 2030 in relevant settings. That posture gives financial regulators a strong basis to require crypto products, custodians, collateral programs, and market infrastructure to file ACP evidence rather than rely on generic disclosure.

Policy surface	Current posture and ACP application
PQC policy	ASD/ACSC is explicit that DH, ECDH, ECDSA, and RSA are not approved beyond 2030 in relevant settings; ML-DSA-65/87 and ML-KEM-768/1024 are recognized, with higher assurance parameters preferred for new cryptographic equipment by 2030.
Digital-asset policy	APRA, ASIC, RBA, and Treasury provide operational-risk, digital-asset, payments, collateral, licensing, custody, and market-infrastructure hooks.
Current gap	Public materials do not yet impose a crypto-asset transfer-authority gate equivalent to ACP for digital-asset licensing, custody, collateral, payments, or market-infrastructure use.
Risk to investors	Australian investors can receive digital-asset access through licensed products, custodians, collateral arrangements, or payment services even though vulnerable public-key cryptography is already time-limited in other assurance contexts. Without an asset-level gate, public holders may bear the loss from late migration, collateral rejection, custody interruption, or a disorderly market response after the 2030 assurance line approaches.
Framework use	Use ACP in ASIC digital-asset licensing, APRA CPS 234/CPS 230 assessments, custody and outsourcing reviews, RBA payments/FMI work, and collateral eligibility. Because BTX's reviewed active ML-DSA parameter is ML-DSA-44, Australian application should separate NIST-family transfer authority from ASD-preferred parameter posture and require an upgrade or equivalence plan where domestic assurance levels demand it.

7.12 International Regulatory Finding

This specimen is the written output of the preceding cross-jurisdictional analysis. It gives a non-U.S. regulator a neutral way to adopt the framework without importing U.S. statutory labels or implying that BTX has been selected for investment, listing, or legal-status purposes. The regulator should replace the bracketed domestic hook with its own authority and attach any jurisdiction-specific legal memorandum, but the technical benchmark, claim-boundary discipline, evidence package, and consequence structure should remain intact.

EXAMPLE OUTPUT		SPECIMEN RECORD
Cross-Jurisdiction Determination		
Finding	The framework is U.S.-originated but jurisdiction-neutral in execution.	
Domestic hook	[product authorization / trading venue / custody / collateral / payment / clearing / operational resilience / disclosure authority]	
Technical benchmark	BTX's regulator-facing reference functions or a documented functionally equivalent post-quantum transfer-authority profile.	
Required adaptation	Replace SEC/CFTC legal citations with domestic authority; preserve ACP classification, claim-boundary discipline, CBOM, vectors, audit evidence, custody tests, and market-operational consequences.	
Regulatory consequence	[approval / conditional approval / disclosure / restriction / haircut / halt / wind-down / enforcement referral]	

Specimen 8: Jurisdiction-neutral determination language for regulators adopting the framework outside the United States.

8 Transition Program for Existing Assets

8.1 Program Objective

The BTX reference model becomes regulator-useful when it is treated as a transition framework, not merely a classification label. BTX shows how an existing Bitcoin-equivalent asset can preserve its commodity-style profile while replacing the cryptographic authority that moves value. A regulator can use that pattern to require any covered asset to document how it will preserve lawful market functions, replace vulnerable transfer authority, shut down nonconforming paths, support custodians and market operators, and protect retail users during migration.

The transition program has two lanes. The asset-lane controls protocol, account, validator, bridge, governance, wallet, and settlement mechanics. The market-lane controls listing, custody, ETP, index, clearing, collateral, margin, customer disclosure, and incident response. A filer cannot complete the transition by addressing only one lane. A technically upgraded asset that cannot be custodyed, withdrawn, valued, or liquidated safely remains operationally deficient; a hardened custodian cannot make a vulnerable base asset ACP-Q3 by itself.

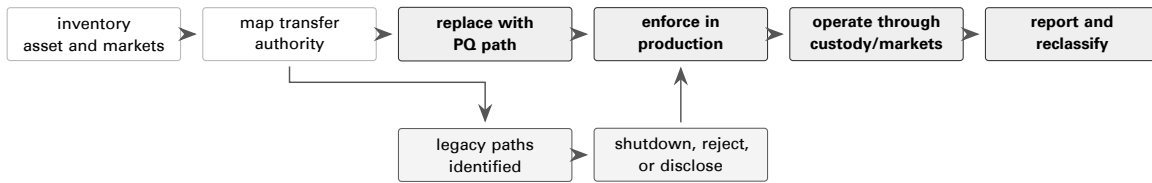


Figure 6: BTX-derived transition flow for existing assets. The migration is complete only when production enforcement, operational readiness, and ongoing reporting align.

8.2 Transition Phases

The phase table is the regulator’s migration ladder for existing assets. It prevents a filer from moving directly from a roadmap or testnet demonstration to ACP-Q3 treatment without proving production activation, legacy-path shutdown, wallet and custody operability, and continuing supervision. Each phase should be used as a milestone in product conditions, custody remediation plans, collateral policies, or market-operator admission files.

Phase	Objective	Required work	ACP result
0	Baseline and exposure inventory	Fix claim boundary; identify asset properties, products, custody uses, market uses, public-key exposure, bridge/wrapper exposure, and all transfer-authority paths.	ACP-U until complete; normally ACP-Q0 if vulnerable paths remain.
1	Public transition design	Publish target PQ transfer model, standards mapping, governance path, activation plan, wallet/custody plan, customer migration plan, and negative-test plan.	ACP-Q1 if credible, dated, and testable.
2	Production-like pilot	Demonstrate signing, validation, rejection, wallet, custody, exchange, bridge, index, and recovery flows in production-like conditions.	ACP-Q1 remains; ACP-Q2 possible only if hybrid-active production evidence exists.
3	Production activation	Activate post-quantum or hybrid transfer authority under production rules with downgrade and replay resistance.	ACP-Q2 for hybrid; ACP-Q3 only when ordinary transfer no longer needs vulnerable signatures.
4	Legacy shutdown and migration	Sweep exposed balances, rotate keys, disable old paths, restrict unsupported wallets, update products, and publish residual-exposure metrics.	ACP-Q3 possible when legacy production movement is shut down or outside the claim boundary.

Phase	Objective	Required work	ACP result
5	Continuing supervision	Maintain CBOM, vectors, custody tests, market reporting, incident playbooks, release attestations, and material-change notices.	ACP maintained, downgraded, or suspended based on evidence.

8.3 Model Agency Text

The following inserts are model operative clauses. They are written in agency language so that staff can adapt them into rule text, interpretive guidance, exchange rules, product orders, custody conditions, or collateral eligibility procedures. They should be used with the definitions, compliance dates, and evidence requirements above; standing alone, they are not enough to complete a conformance program.

Reference-Asset Insert

The Commission identifies BTX's regulator-facing reference profile as the benchmark for post-quantum digital commodity conformance and as the worked example for transition from quantum-vulnerable transfer authority to production-enforced post-quantum transfer authority. A covered asset that seeks ACP-Q3 treatment must either conform to those BTX reference functions or demonstrate functionally equivalent or superior production transfer authority under NIST-standardized post-quantum digital signatures. Functional equivalence is measured by production enforcement, rejection of nonconforming transfer paths, wallet and custody operability, market readiness, reproducible evidence, and accurate disclosure.

Transition Filing Requirement

A covered filer seeking approval, continued listing, custody onboarding, collateral eligibility, clearing treatment, or retail product exposure for an ACP-Q0, ACP-Q1, or ACP-Q2 asset must file a transition plan that identifies the target ACP, migration phases, transfer-authority map, legacy-path shutdown plan, custody readiness plan, market-operator readiness plan, customer notice plan, and milestone consequences. A transition plan does not authorize post-quantum marketing unless the evidence supports the claimed ACP.

8.4 Application to Bitcoin, Ethereum, Solana, HYPE, and BTX

This table applies the same ACP method to prominent assets and products to show that the framework is evidence-based rather than brand-based. Bitcoin, Ethereum, Solana, and HYPE illustrate different vulnerable transfer-authority surfaces that require transition evidence before favorable treatment. BTX is included as the reference asset because it supplies the worked production example; other assets can qualify only by showing functional equivalence or superiority for the same regulatory functions.

Asset	Initial ACP posture	Transition issue	Required regulatory treatment
Bitcoin	ACP-Q0 unless a production PQ spend path and transition plan are adopted.	Ordinary transfer authority depends on secp256k1 signatures; exposed public keys and reused address practices create measurable migration exposure.	Require asset-layer ACP, exposed-key metrics, wallet/custody migration plan, production enforcement proposal, ETP risk-factor update, and collateral/custody limitations until transition evidence improves.
Ethereum	ACP-Q0 or ACP-Q1 depending on credible transition evidence.	Account keys, validator keys, staking withdrawal controls, smart-contract admin keys, bridges, rollups, and governance mechanisms create multiple transfer-authority surfaces.	Require network-layer and intermediary-layer CBOM, validator and account migration evidence, smart-contract and bridge exposure mapping, custody transfer testing, and milestone-based product conditions.
Solana	ACP-Q0 or ACP-Q1 depending on credible transition evidence.	Ordinary transactions rely on signer-provided Ed25519 signatures; validator, vote, stake, program-upgrade, bridge, custody, wallet, RPC, and indexer dependencies create additional transfer-authority and market-operation surfaces.	Require asset-specific ACP, account and signer exposure mapping, validator/staking and program-authority CBOM, wallet and custody migration tests, RPC/indexer readiness evidence, and product or listing conditions until production PQ transfer authority is proven.
HYPE	ACP-Q0 or ACP-Q1 depending on protocol evidence.	Retail-facing Hyperliquid ETP filings already acknowledge quantum and cryptography risk, but disclosure does not prove validator, governance, bridge, custody, or market-operation conformance.	Require asset-specific ACP, validator/-governance/key exposure analysis, on-chain market stress plan, ETP disclosure tied to ACP class, and exchange/custody readiness evidence.
BTX	ACP-Q3 reference profile, subject to evidence maintenance.	BTX supplies the worked example: preserve Bitcoin-equivalent commodity features while replacing vulnerable transfer authority with production-enforced post-quantum authorization.	Use BTX as the reference asset and compare other assets to the generic functions it demonstrates: production enforcement, legacy-path shutdown, wallet/custody operability, market readiness, and evidence discipline.

9 Audit, Test-Vector, and Evidence Program

This section converts BTX's reference functions into an examinable certification method. The program defines the evidence record a filer provides, the audit domains an examiner applies, the vector manifests required for repeatable testing, the CBOM content required for cryptographic inventory review, and the pass/fail determinations used to assign an ACP class.

For regulatory use, the desk-review function and the filer-evidence function are distinct. The desk-review record maps source materials, architecture, regulatory hooks, filing evidence, and BTX code exhibits to a supervisory framework. External laboratory attestations, ACVP/CAVP records, CMVP/FIPS 140-3 certificates where claimed, custody test reports, and independent cryptographic audits are the formal evidence inputs a filer supplies to make a public conformance claim enforceable.

9.1 Claim Boundary and Certification Record

A conformance review begins by fixing the claim boundary. The boundary identifies the asset, network, release, commit hash, activation height or activation rule, covered product, covered custody systems, covered wallet flows, and excluded non-production modes. A claim that mixes production and test modes, omits activation state, or relies on a repository different from the deployed binary fails before reaching technical review.

The certification record then links that boundary to six exhibits. The ACP claim explains the requested ACP-Q0, ACP-Q1, ACP-Q2, or ACP-Q3 classification by mapping production transfer authority to BTX's reference functions. The CBOM inventories protocol, wallet, custody, build, vendor, HSM, MPC, API, transport, governance, and recovery dependencies. The vector package supplies signed algorithm, protocol, negative, replay, downgrade, wallet, custody, and interoperability vectors. The attestation record supplies independent cryptographic audit evidence, laboratory vector evidence, module-validation evidence where FIPS 140-3 is claimed, and custody or wallet interoperability evidence. The operational record supplies activation status, node/miner/validator adoption, exchange and custodian readiness, supply and liquidity data, fee and size effects, orphan/reorg data, and emergency CRQC playbooks. The disclosure record supplies public-facing language, marketing controls, risk factors, limitation statements, and regulator-facing certifications.

9.2 Audit Domains and Determinations

The audit-domain table is the examiner's scope control. It tells the reviewer which domains must be inspected before any ACP classification can be assigned and separates the evidence needed for protocol behavior, cryptographic claims, implementation security, and market operation. A conditional or incomplete domain may support transition treatment, but it should not be converted into ACP-Q3 unless the defect is outside the claim boundary or has been remediated and retested.

ID	Domain	Required evidence	Pass/fail determination
D1	Claim boundary	Asset, network, release, commit, activation rule, products, custody systems, and excluded modes.	Pass if the reviewed system is uniquely identified. Fail if production and non-production scope are mixed or unverifiable.
D2	ACP classification	Transfer-authority analysis mapped to ACP-Q0 through ACP-Q3.	Pass if the claimed ACP matches actual production transfer authority. Fail if a quantum-vulnerable ordinary transfer path is presented as ACP-Q3.
D3	Standards mapping	FIPS 203, FIPS 204, FIPS 205, successor standards, parameter sets, and roles.	Pass if every PQC claim maps to a named standard, parameter set, and use. Fail if an unsupported algorithm is marketed as NIST-approved PQC.
D4	Algorithm conformance	ACVP/CAVP evidence or laboratory KAT attestation for the exact implementation and parameter set.	Pass if vectors match the claimed implementation. Fail if claimed algorithms lack reproducible validation evidence.
D5	Key generation and entropy	RNG/DRBG design, entropy source, deterministic options, derivation rules, and key lifecycle.	Pass if keys are generated and controlled under documented processes. Fail for repeated secrets, weak entropy, or undocumented derivation.

ID	Domain	Required evidence	Pass/fail determination
D6	Domain separation and serialization	Address format, authorization encoding, message preimage, transaction digest, commitment proof, and context rules.	Pass if signatures bind the intended domain and serialized object. Fail for ambiguous encodings, malleable commitments, or cross-domain acceptance.
D7	Consensus enforcement	Block, transaction, authorization, policy, fork, activation, and rejection rules.	Pass if production validation enforces the claimed transfer model. Fail if production consensus accepts a prohibited legacy or downgrade path.
D8	Wallet, custody, and transaction support	Wallet defaults, custody/API fields, hardware-wallet paths, sweep, withdrawal, recovery, and wind-down tests.	Pass if covered custodians can sign, verify, withdraw, recover, and wind down under the claimed model. Fail if customer transfer capability is unavailable.
D9	Negative vectors	Invalid signature, malformed key, wrong commitment, wrong context, replay, legacy signature, downgrade, and malformed authorization vectors.	Pass if all invalid cases are rejected with documented errors. Fail if any mandatory invalid vector is accepted.
D10	Transport and KEM scope	ML-KEM or hybrid key-establishment evidence for transport paths where claimed.	Pass if transport claims are separated from ownership claims and tested. Fail if KEM evidence is used to imply transfer-authority conformance.
D11	Implementation hardening	Constant-time review, memory safety, side-channel review, fault handling, unsafe optimization review, and dependency analysis.	Pass if no critical implementation defect remains open. Fail for exploitable timing, memory, downgrade, or fault-injection findings.
D12	Build and provenance	Source tag, build recipe, compiler, dependency lockfiles, signed binaries, reproducibility logs, and SBOM linkage.	Pass if binaries and deployed artifacts trace to reviewed source. Fail if production artifacts cannot be tied to the reviewed release.
D13	Network activation and economics	Activation height, node/miner/validator adoption, fee impact, size impact, storage growth, liquidity, and market-readiness data.	Pass if production status and market operation support the claimed ACP. Fail if ACP-Q3 is claimed before production activation.
D14	Fork, replay, and CRQC response	Fork playbook, replay controls, emergency triggers, customer notices, withdrawal procedures, and governance path.	Pass if response procedures preserve customer and market integrity. Fail if no executable response path exists for material cryptographic failure.
D15	Disclosure and marketing controls	Risk factors, public claims, limitation statements, product documents, and sales controls.	Pass if statements match the evidence level. Fail for unsupported “quantum-safe,” “NIST-certified asset,” or equivalent claims.
D16	Third-party dependencies	HSM, MPC, wallet, API, oracle, bridge, explorer, exchange, custodian, and vendor CBOM entries.	Pass if external dependencies are inventoried and covered by attestations. Fail if a material unreviewed dependency controls transfer, custody, or recovery.

9.3 NIST/FIPS Claim Levels

The framework separates three evidence levels because the legal meaning of a PQC claim depends on the exact boundary. Specification conformance means the implementation follows an algorithm specified in FIPS 203, FIPS 204, or FIPS 205. Algorithm validation means the exact implementation and parameter set have ACVP/CAVP evidence or equivalent laboratory attestation. Module validation means the defined cryptographic module boundary has CMVP/FIPS 140-3 validation, including operating environment, security policy, approved-mode configuration, and certificate status.

Evidence level	Meaning	Permitted claim
Specification conformance	The implementation follows an algorithm specified in FIPS 203, FIPS 204, or FIPS 205.	Uses a NIST-specified algorithm, limited to the stated role and parameter set.
Algorithm validation	The exact implementation has ACVP/CAVP evidence or laboratory vector evidence for the claimed algorithm and parameter set.	Validated or independently tested implementation for the stated algorithm, parameter set, and version.
Module validation	The defined cryptographic module has CMVP/-FIPS 140-3 validation, including boundary, operating environment, security policy, and approved-mode status.	FIPS-validated module, limited to the certified boundary and approved-mode conditions.

BTX's role in this framework rests on the production transfer-authority architecture documented in the code exhibits: production-enforced post-quantum transfer authority, PQ signature authorization, legacy-path rejection, and wallet/custody operability. If a filer makes a stronger laboratory or FIPS-module claim, that stronger claim attaches only when the filer supplies the corresponding evidence for the exact implementation and boundary.

9.4 Required Vector Package

The vector record is organized by claim, not by implementation component. Each vector family identifies the claim it supports, the network or activation state assumed, the expected result, and the negative cases required to prove downgrade resistance. The filer supplies a signed `vectors/manifest.json` and one suite manifest per test family. Each manifest entry contains the vector identifier, requirement identifier, algorithm or protocol surface, parameter set, implementation hash, input encoding, expected output, expected rejection code for negative cases, execution command, execution environment, source reference, artifact hash, and attesting party.

Suite	Name	Required contents
V1	ML-DSA	KeyGen, sigGen, sigVer, malformed signature, malformed public key, wrong message, wrong context, and boundary-size vectors for every supported ML-DSA mode.
V2	SLH-DSA	KeyGen, sigGen, sigVer, FIPS-205 activation behavior, legacy-mode rejection where applicable, malformed signature, malformed public key, wrong message, and wrong context vectors.
V3	ML-KEM	KeyGen, encapsulation, decapsulation, decapsulation failure, transcript binding, rekeying, and downgrade-negative vectors for transport paths where claimed.
V4	Address and commitment	Address encoding, transfer-authority commitment, BTX's P2MR commitment where the filer is BTX, or a functionally equivalent commitment for another asset, authorization proof, and wrong-commitment rejection vectors.
V5	Transaction and custody workflow	Signature preimage, transaction-support fields, signing flow, finalization flow, fee and size boundary cases, and multi-input/multi-output vectors.
V6	Block and consensus	Valid block, invalid nonconforming standard-transfer path, invalid legacy signature path, activation-boundary block, reorg boundary, policy rejection, and consensus rejection vectors.

Suite	Name	Required contents
V7	Negative and downgrade	Replay, cross-network replay, wrong chain identifier, wrong activation state, unsupported algorithm, legacy-signature fallback, malformed authorization data, and parser-confusion vectors.
V8	Wallet and custody	Address generation, deposit, withdrawal, sweep, recovery, rotation, hardware-wallet signing, MPC/HSM signing, API signing, and wind-down transfer vectors.
V9	Interoperability	Explorer, exchange, custodian, indexer, node RPC, light-client, and monitoring samples proving consistent parsing, verification, and rejection behavior.

9.5 CBOM Content

The CBOM is a maintained inventory, not a narrative appendix. It is filed in machine-readable form and summarized for examiner review. Its purpose is to stop a filer from making an asset-level PQC claim while omitting a wallet, custody, bridge, governance, or vendor dependency that still controls value through quantum-vulnerable keys.

CBOM category	Required fields
Project identity	Legal filer, asset, network, product, repository, commit, release tag, binary hash, activation status, and responsible control owner.
Cryptographic algorithms	Algorithm name, standard, parameter set, role, implementation module, approved mode, fallback behavior, and claimed evidence level.
Keys and signatures	Key type, generation process, derivation path, public-key exposure state, signing surface, verification surface, rotation process, backup process, and destruction process.
Consensus and protocol	Address formats, authorization rules, transaction digest rules, block rules, activation flags, fork controls, replay controls, and downgrade controls.
Libraries and modules	Package, version, commit, maintainer, license, build options, validation certificate where claimed, known vulnerabilities, and patch status.
Custody systems	Wallet, HSM, MPC, hardware signer, approval workflow, withdrawal workflow, recovery workflow, vendor, and attestation reference.
Transport and APIs	KEM or hybrid transport use, TLS or session layer, RPC/API authentication, admin access, exchange connectivity, and monitoring channels.
Build and release	Compiler, OS, dependency lockfile, build recipe, reproducibility status, signed binary evidence, SBOM cross-reference, and release authority.
Operations and incident response	Monitoring, alerting, emergency CRQC triggers, fork procedure, customer notice process, wind-down process, and tabletop evidence.
Residual exposure	Legacy balances, exposed public keys, unsupported wallets, bridge or wrapper dependencies, governance keys, validator keys, and mitigation status.

9.6 Pass/Fail and ACP Outcomes

Each mandatory audit domain receives **Pass** or **Fail**. A domain may be marked **Not Applicable** only when the claim boundary excludes that surface and the exclusion is explicit in the filing. A critical defect produces a failed certification result. A major defect produces a failed certification result until remediated and retested. A minor observation is recorded without changing a pass result.

Outcome	Determination
ACP-Q3 Certified	Production ordinary transfer authority uses NIST-approved PQC digital signatures without requiring a quantum-vulnerable signature; all applicable domains pass; required vector suites pass; CBOM is complete; external attestations support the claim.
ACP-Q2 Certified	Production transfer authority is hybrid-active and downgrade-resistant; PQC and classical paths are bound together; all applicable domains pass; disclosures identify remaining classical dependency.
ACP-Q1 Transition Accepted	The asset remains quantum-vulnerable in production but has credible public specifications, test evidence, governance path, and dated milestones. PQC marketing claims are prohibited except as transition-status descriptions.
ACP-Q0 PQC Status Denied	Ordinary transfer authority depends on a quantum-vulnerable signature path, the evidence package is incomplete, or the claimed PQC transfer model fails a mandatory domain.

9.7 Attestation Text

Use these three attestations in sequence: filer, external laboratory, then regulator. The regulator should not issue an ACP determination until the filer and laboratory records identify the same claim boundary and the same implementation evidence. These specimen records are designed as copy-ready filing outputs, but each must be supported by the underlying CBOM, vector manifests, audit reports, custody tests, operational evidence, and disclosure materials.

The sequence matters. The filer attestation creates officer accountability for completeness; the laboratory attestation binds testing to a repository, commit, binary, module boundary, and vector manifest; the regulatory determination records the examiner's classification and limitations. None of the specimens is a standalone certification, and none should be used to imply NIST, FIPS 140-3, laboratory, or regulator approval beyond the evidence actually obtained.

EXAMPLE OUTPUT		SPECIMEN RECORD
Filer Evidence Attestation		
Filer	[legal name / responsible officer]	
Asset or product	[asset/product]	
Claim boundary	[network, release, activation state, custody systems, products, exclusions]	
Certification	The undersigned certifies that the CBOM, vector manifests, audit reports, custody test records, and operational evidence submitted for the stated claim boundary are complete and accurate in all material respects.	
Dependency statement	The filer has identified all material cryptographic dependencies used for transfer authority, custody, recovery, transport, governance, and market operation.	
Omission statement	The filer has not omitted any known production path that permits ordinary asset transfer outside the claimed ACP classification.	
Signature	[name / title / date]	

Specimen 9: Filer attestation used to certify the completeness of the CBOM, vector manifests, audit record, custody evidence, and claim boundary.

EXAMPLE OUTPUT		SPECIMEN RECORD
External Laboratory Attestation		
Laboratory	[laboratory]	
Implementation	repository [repository]; commit [commit]; release [release]; binary hash [hash]; module boundary [boundary]	
Vector manifest	vectors/manifest.json	
Test result	The laboratory executed the identified vector suites, verified the expected pass and rejection results, and confirmed the evidence references listed in this attestation.	
Scope limitation	This attestation covers only the identified implementation, parameter sets, operating environment, and test date.	
Authorized signatory	[name / title / date]	

Specimen 10: External laboratory attestation used to bind test-vector execution and audit evidence to a specific repository, commit, binary, and operating boundary.

EXAMPLE OUTPUT		SPECIMEN RECORD
Regulatory Determination		
Determining authority	[regulator/examiner]	
Asset or product	[asset/product]	
Evidence reviewed	desk-review record; external attestations; CBOM; vector manifests; operational evidence; disclosure materials	
Classification	[ACP-Q0/Q1/Q2/Q3] for the stated claim boundary	
Limitation	This determination is not an endorsement of investment merit, market value, or absence of operational risk.	
Date	[date]	

Specimen 11: Regulatory determination record used to assign an ACP classification after the filer and laboratory evidence has been reconciled.

9.8 Regulatory Application to Any Project

Regulators apply this framework by normalizing any project into six review surfaces: transfer authority, consensus or settlement, custody and signing, transport and APIs, governance and upgrade control, and product disclosure. The same method applies to Bitcoin-style output-accounting assets, account-based networks, proof-of-stake systems, app-chains, bridges, wrapped assets, wallets, custodians, exchanges, and ETP sponsors.

The examiner first fixes the claim boundary, then verifies the CBOM, then runs the audit domains, then checks the vector manifests and external attestations, then assigns the ACP outcome. A project qualifies for ACP-Q3 only when production ordinary transfer authority is post-quantum under the claimed standard and the complete evidence record passes. A project with strong custody controls but a quantum-vulnerable base protocol does not qualify as ACP-Q3 for the asset; it may receive a separate custody-system determination within its own claim boundary.



Figure 7: Evidence pipeline for post-quantum digital-asset conformance.

10 Custodian and Wallet Provider Obligations

Custody controls are a separate examination surface from asset-level conformance. A custodian may operate a strong HSM, MPC, hardware-signer, approval, and monitoring stack while the underlying asset remains quantum-vulnerable. The custody claim must therefore be scoped precisely: it identifies what the custodian or wallet provider controls, what the base asset controls, what vendors control, what customers control, and what residual legacy exposure remains.

Model Custodian and Wallet Provider Rule

A covered custodian, hosted wallet provider, wallet-as-a-service provider, broker, trust company, FCM, DCO, clearing member, fund custodian, or other intermediary that holds, controls, signs for, administers, or materially facilitates transfers of customer digital assets must maintain a custody-specific ACP record, custody CBOM, customer-asset segregation record, residual legacy exposure report, and tested transfer capability. A covered person may not represent that custody is post-quantum, quantum-safe, quantum-resistant, or functionally equivalent to BTX's reference functions unless the statement is accurate for the stated claim boundary and does not imply that an ACP-Q0, ACP-Q1, or ACP-Q2 base asset has become ACP-Q3.

10.1 Custody Claim Boundary

The custody claim boundary is the first examiner control. It prevents a filer from using a hardened custody stack to overstate the post-quantum status of the underlying asset, or from using an asset-level ACP claim to hide vulnerable custody operations. The boundary must be written at the level of legal entity, asset, network, account, wallet system, signing workflow, customer class, vendor dependency, and product use.

Boundary surface	Required statement	Regulatory consequence
Base asset	Asset, network, fork, release, activation status, ACP class, bridge or wrapper form, and residual legacy transfer paths.	A custody-system claim cannot upgrade the base asset ACP. ACP-Q3 treatment requires asset-level evidence.
Custody system	Wallet architecture, hot/warm/cold tiers, HSM, MPC, hardware signer, quorum policy, approval workflow, recovery workflow, and monitoring scope.	Custody approval is limited to the identified systems and customer asset classes.
Wallet provider role	State whether the provider controls keys, co-signs, supplies software only, supplies hardware only, provides wallet-as-a-service, or provides transaction construction without custody.	Hosted and co-signing providers receive custody-style obligations; software-only providers receive CBOM, disclosure, update, and anti-downgrade obligations.
Customer property scope	Identify omnibus wallets, segregated wallets, sub-ledgers, customer-specific addresses, internal transfers, staking/lending exceptions, and proprietary asset separation.	Ambiguous or commingled scope requires remediation before new custody onboarding.
Excluded systems	Identify non-production tools, unsupported wallets, legacy addresses, manual break-glass procedures, third-party bridges, unsupported hardware, and discontinued assets.	Exclusions must not be usable to move customer value outside the filed controls.

10.2 Custody CBOM

A custody CBOM must inventory every cryptographic component that can authorize, block, recover, delay, approve, or redirect customer asset transfers. The CBOM must be filed in machine-readable form and summarized for examination.

Component	CBOM fields	Examiner test
HSM	Vendor, model, firmware, module boundary, algorithm set, approved mode, key export policy, admin roles, audit logs, patch status, and validation certificate where claimed.	Confirm that the HSM can produce or protect the claimed signature workflow and that no quantum-vulnerable fallback can move customer value outside policy.
MPC system	Protocol, threshold, participant roles, share generation, share refresh, transcript binding, signing policy, recovery quorum, vendor services, and failure assumptions.	Execute normal signing, failed quorum, share recovery, participant removal, and downgrade-negative tests.
Hardware signer	Device, secure element, firmware signing, supported assets, supported PQ or hybrid paths, display verification, transaction parser, backup method, and update channel.	Confirm that the device signs the intended transaction and rejects wrong-chain, wrong-commitment, legacy, replay, and malformed transaction cases.
Wallet software	Repository, release, binary hash, address policy, account or transaction-support model, API handling, fee policy, public-key exposure rules, and customer warnings.	Reproduce deposit, withdrawal, sweep, recovery, and rejected downgrade vectors.
Approval system	Human approvals, automated limits, velocity rules, customer authentication, role separation, freeze authority, release authority, and emergency override controls.	Confirm dual-control, least privilege, time-lock, exception logging, and release-after-freeze evidence.
Recovery system	Seed, shard, backup, legal recovery, disaster recovery, alternate custodian, customer-directed transfer, key destruction, and wind-down mechanics.	Demonstrate recovery without commingling customer assets or bypassing the claimed transfer-authority model.

10.3 Customer Asset Segregation

Customer asset segregation has both legal and cryptographic dimensions. A custodian must segregate customer property from proprietary assets, and must also segregate signing authority, approval authority, address pools, migration queues, and residual legacy exposure so that a weakness in one class of assets does not silently contaminate another.

Customer Asset Segregation Rule

A covered custodian must maintain records and controls reasonably designed to demonstrate that customer digital assets are not property of the custodian, are not pledged or rehypothecated absent express authority, are not commingled with proprietary assets, and can be transferred, swept, recovered, frozen, released, or wound down according to customer entitlements. Segregation records must reconcile on-chain balances, omnibus wallet balances, internal sub-ledgers, pending withdrawals, fees, forks, airdrops, bridge claims, and residual legacy exposure.

Control	Required evidence	Failure consequence
Legal segregation	Customer agreements, account terms, trust or custodial records, lien restrictions, and insolvency treatment analysis.	Corrective disclosure, custody restriction, or wind-down plan.
Ledger segregation	Daily reconciliation between customer sub-ledger, omnibus addresses, pending transfers, fees, and corporate treasury.	New deposits restricted until reconciliation exceptions are cured.

Control	Required evidence	Failure consequence
Key segregation	Separate key material, policies, quorums, and approval roles for customer assets, house assets, test assets, and vendor operations.	Critical finding if one key path can transfer both customer and proprietary value without separate authorization.
Migration segregation	Separate queues for ACP-Q0, ACP-Q1, ACP-Q2, and ACP-Q3 assets; separate sweep plans for exposed legacy addresses and unexposed addresses.	ACP downgrade review if migration records blur customer priority or exposure class.
Exception handling	Documented treatment for forks, airdrops, sanctions holds, disputed ownership, court orders, network halts, and CRQC events.	Freeze-release test required before continued custody approval.

10.4 Required Operational Tests

Operational readiness is demonstrated by execution evidence, not policy text. Each test must identify the asset, ACP class, wallet tier, customer class, signer stack, approver roles, transaction identifiers or simulator evidence, expected result, exception handling, customer notice, and remediation owner.

Test	Minimum execution	Pass condition
Withdrawal	Customer-initiated transfer from each supported custody tier, including normal, high-value, fee-stress, and delayed-review cases.	Transfer completes through the approved path; customer record, on-chain record, approval log, and notice log reconcile.
Sweep	Batch movement from hot or legacy addresses into approved destination addresses, including dust, fee, stuck-transaction, and partial-failure cases.	Sweep preserves customer entitlements, avoids unnecessary public-key exposure, and rejects non-conforming transfer paths.
Recovery	Lost device, unavailable signer, unavailable MPC participant, corrupted wallet metadata, failed vendor, and disaster-recovery scenarios.	Recovery restores transfer capability without using an undisclosed key path or mixing customer and proprietary assets.
Wind-down	Transfer of all customer assets for an asset, product, or custody relationship to customers or a successor custodian.	The custodian can exit the activity while preserving customer choice, records, tax lots where maintained, and ACP disclosure.
Freeze-release	Sanctions hold, suspected compromise, fork/replay event, CRQC trigger, legal process, false positive, and release authorization.	Freeze is scoped, logged, legally supported where required, communicated when required, and released through a tested authority path.
Customer notice	Pre-migration notice, ACP downgrade notice, withdrawal delay notice, incident notice, wind-down notice, and corrective disclosure.	Notice is timely, accurate, product-specific, and does not overstate PQC status or custodian discretion.

10.5 Residual Legacy Exposure

Residual legacy exposure means any customer or operational value that can still be affected by a quantum-vulnerable, nonconforming, unsupported, or untested transfer path. Cold storage does not eliminate residual exposure if the asset-level public key, legacy transfer path, bridge key, governance key, validator key, recovery key, or custody approval key can still authorize value movement under quantum-vulnerable assumptions.

Exposure metric	Required measurement	Threshold action
Legacy customer balances	Value, customer count, address count, product use, public-key exposure state, and age of exposure.	Customer notice and migration plan if exposure is material or exceeds filed milestones.
Exposed public keys	Addresses, accounts, validators, bridge keys, governance keys, custody keys, and reused keys with public keys already revealed.	Accelerated sweep, withdrawal restriction, or heightened disclosure depending on ACP class.
Unsupported wallets	Customer balances or transfer flows requiring software, hardware, authorization formats, bridges, or APIs outside the claim boundary.	No new deposits; supported withdrawal path or wind-down required.
Vendor legacy paths	HSM, MPC, hardware signer, recovery, API, firmware, or approval systems using RSA, ECDSA, EdDSA, Schnorr, ECDH, or similar vulnerable primitives for material control.	CBOM remediation and retest before PQC custody claims may continue.
Bridge or wrapper dependency	Wrapped assets, lockboxes, mint/burn controls, oracle keys, relayer keys, and emergency multi-sigs.	Separate ACP treatment for wrapper and base asset; no ACP-Q3 claim unless both boundaries pass.

10.6 Treatment by Asset ACP

Custody treatment follows the base asset ACP and the custody-system claim boundary. The custodian may impose stronger controls than the table below, but may not apply weaker controls without documented relief and customer-protection analysis.

ACP	Custody posture	Required controls	Customer treatment
ACP-Q0	Legacy exposure	No PQC marketing claim; heightened disclosure; no new omnibus custody after phase-in dates absent relief; withdrawal and wind-down capability required.	Clear notice that ordinary transfer remains quantum-vulnerable; customer-directed withdrawal path maintained where lawful and operationally feasible.
ACP-Q1	Transition-ready	Custody permitted only with dated migration milestones, residual exposure report, quarterly status, and tested sweep/recovery procedures.	Customers receive milestone notices, missed-milestone notices, and migration support.
ACP-Q2	Hybrid-active	Custodian must support the hybrid production path, reject downgrade attempts, maintain annual transfer tests, and disclose remaining classical dependency.	Customers receive accurate hybrid-status disclosure and notice of any downgrade, bypass, or failed vector.
ACP-Q3	Reference-profile equivalent or superior	Eligible for post-quantum custody treatment if custody CBOM, signer tests, segregation, and residual exposure review also pass.	Standard custody disclosures may state ACP-Q3 status within the stated boundary and evidence date.

10.7 Custodian Readiness Flow

The flow below turns custody evidence into a supervisory decision path. It begins with the custody claim boundary and base-asset ACP, then tests the custodian's own signer stack, transfer capability, segregation evidence, and residual exposure. The figure should be read as a process diagram: an approved custody outcome requires both a conforming or conditionally acceptable base asset and operational proof that the custodian can move, recover, segregate, and wind down customer assets under the claimed model.

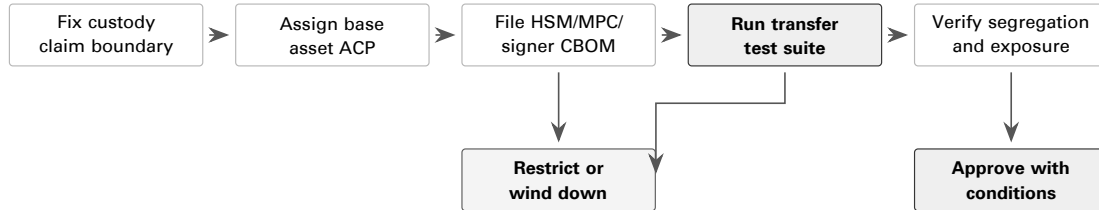


Figure 8: Custodian readiness flow: the custody claim boundary, base asset ACP, signer CBOM, operational tests, segregation record, and residual exposure report determine whether custody is approved, conditioned, restricted, or wound down.

10.8 Custody Examination Checklist and Scorecard

This subsection gives examiners two related instruments. The first table is a checklist for intake and deficiency review; it tells staff what must be present before a custody determination is complete. The second table is the scoring rubric used after the critical gates and operational tests have been reviewed. A high numerical score cannot cure a failed transfer test, a missing CBOM, deficient segregation, or an inability to return customer assets.

	Checklist item	Evidence	Result
☐	Confirm custody claim boundary.	Legal entity, asset list, wallet tiers, vendors, products, exclusions.	Complete / deficient.
☐	Reconcile base asset ACP to custody treatment.	ACP record, residual exposure report, customer disclosures.	Q0 / Q1 / Q2 / Q3 / U.
☐	Review HSM, MPC, and hardware signer CBOM.	Machine-readable CBOM, vendor attestations, firmware and module records.	Complete / deficient.
☐	Execute withdrawal, sweep, recovery, and wind-down tests.	Test logs, transaction evidence, simulator output, approval logs.	Pass / fail / retest.
☐	Verify customer asset segregation.	Ledger reconciliation, address inventory, key policy, proprietary asset separation.	Pass / fail.
☐	Assign custody conditions.	Examiner memo, scorecard, remediation dates, customer notice obligations.	Approved / conditioned / restricted / wind-down.

ID	Domain	Pts	Scoring rule
CS1	Claim boundary	10	Full credit for precise entity, asset, wallet, vendor, customer, product, and exclusion scope.
CS2	Custody CBOM	15	Full credit for complete HSM, MPC, hardware signer, wallet, API, firmware, recovery, and vendor cryptographic inventory.
CS3	Transfer tests	20	Full credit for passed withdrawal, sweep, recovery, wind-down, freeze-release, and customer-notice tests across material asset classes.
CS4	Segregation	15	Full credit for legal, ledger, key, migration, and exception segregation with daily reconciliation.
CS5	Residual exposure	15	Full credit for measured legacy balances, exposed public keys, unsupported wallets, bridge dependencies, and mitigation milestones.

ID	Domain	Pts	Scoring rule
CS6	ACP treatment controls	10	Full credit when onboarding, disclosures, limits, and wind-down plans match ACP-Q0, Q1, Q2, and Q3 treatment rules.
CS7	Incident and notice readiness	10	Full credit for current CRQC, fork, replay, custody compromise, freeze-release, and customer notice playbooks.
CS8	Governance and attestation	5	Full credit for officer certification, risk-committee reporting, material-change notice, and independent review cadence.
Total		100	Score cannot cure a critical failure.

EXAMPLE OUTPUT		SPECIMEN RECORD
Custody Determination		
Determining authority	[regulator/examiner]	
Custodian or wallet provider	[custodian/wallet provider]	
Scope	[asset/product/customer class]	
Evidence reviewed	custody claim boundary; base asset ACP; custody CBOM; operational test record; customer asset segregation evidence; residual legacy exposure report; customer notice controls; management attestation	
Determination	[approved / approved with conditions / restricted / required to wind down]	
Effective date	[date]	

Specimen 12: Custody determination specimen used to document the regulator's decision for a custodian, wallet provider, customer class, or asset scope.

11 Market, Exchange, Clearing, and Product Operator Obligations

Market operators translate ACP status into trading, clearing, margin, product, index, and disclosure consequences. A post-quantum framework is incomplete if it classifies an asset but permits the same listing, collateral, index, custody, and retail product treatment after a material transfer-authority defect is identified.

For purposes of this section, a market operator includes a national securities exchange, alternative trading system, crypto trading platform, DCM, SEF, DCO, FCM, clearing member, ETP sponsor, fund adviser, index administrator, oracle operator, calculation agent, benchmark contributor, issuer, or non-US market infrastructure provider that lists, admits, trades, clears, margins, values, references, custodies, or distributes exposure to a covered digital asset.

Market Operator Control Rule

A covered market operator may not list, admit to trading, clear, margin, include in an index, use as a settlement reference, accept as collateral, or sponsor retail product exposure to a covered digital asset unless it maintains a written market-operator control file. The file must include the asset's ACP classification, claim boundary, CBOM summary, custody and settlement analysis, fork and replay playbook, oracle or index dependency map, margin and collateral treatment, halt and delisting triggers, quarterly reporting controls, and ACP consequence mapping.

BTX supplies the reference implementation for ACP-Q3 review. Other assets may qualify only by showing functionally equivalent or superior production transfer authority, downgrade resistance, wallet and custody operability, and evidence quality. A non-BTX asset cannot receive market-operator treatment equivalent to BTX merely because it discloses quantum risk or has a roadmap.

11.1 Market Operator Control File

The control file is the market operator's standing record for an admitted asset or product. It should be assembled before listing, clearing, margin, index inclusion, or sponsored retail exposure, and then maintained as conditions change. The table identifies the file elements that let an operator connect the asset's ACP posture to trading admission, surveillance intensity, collateral treatment, disclosure updates, and halt or wind-down triggers.

Control file element	Required content	Operator use
ACP record	Current ACP class, claim boundary, examiner date, evidence references, conditions, and next review date.	Determines admission, continued listing, collateral eligibility, product disclosure, and surveillance intensity.
Transfer-authority map	Protocol signing paths, validator or governance keys, bridge keys, custody keys, recovery keys, and exposed public-key inventory.	Identifies whether production value can move through a quantum-vulnerable or nonconforming path.
Custody and settlement file	Deposit, withdrawal, sweep, creation, redemption, liquidation, recovery, and wind-down tests.	Confirms that customers, clearing members, custodians, and product sponsors can move assets under stressed conditions.
Fork and replay playbook	Fork classification, replay protection, chain-selection criteria, wrapped-asset treatment, customer notices, and trading controls.	Supports orderly markets when competing ledgers, migrations, or emergency upgrades affect settlement finality.
Oracle and index map	Source venues, data vendors, calculation agent, oracle keys, methodology, disruption rules, stale-price rules, and fallback sources.	Protects NAV, settlement prices, liquidations, and benchmark references from cryptographic or venue-level failures.
Margin and collateral file	Haircuts, concentration limits, liquidation venue analysis, eligibility limits, default-management treatment, and ACP add-ons.	Prevents a weak ACP asset from transmitting transfer-authority risk into clearing or customer margin.
Disclosure and reporting record	Public risk factors, website disclosures, product documents, quarterly reports, incident notices, and officer attestations.	Aligns market treatment with the evidence actually available to regulators and investors.

11.2 Listing and Admission Review

A listing or admission review must occur before a covered asset is listed for spot trading, included in a product, accepted as collateral, used as a derivative underlier, used as a deliverable, admitted to clearing, added to an index, or relied on for creations and redemptions. The review is not limited to investment merit or liquidity. It determines whether the asset's production transfer-authority model, custody pathways, fork behavior, replay controls, and index dependencies can support an orderly market.

Admission gate	Required evidence	Deficiency consequence
ACP classification	ACP-Q0, Q1, Q2, Q3, or ACP-U determination with claim boundary and evidence date.	ACP-U is treated as ACP-Q0 for admission until cured. Unsupported ACP-Q3 claims are rejected.
Transfer authority	Production validation rules, signature paths, public-key exposure, validator or governance key map, and downgrade tests.	No ACP-Q3 treatment if ordinary transfer can use an unmitigated quantum-vulnerable path.
Custody readiness	Deposit, withdrawal, sweep, recovery, wind-down, creation, redemption, and liquidation tests.	Admission is conditioned, delayed, or denied if assets cannot be moved reliably under the claimed model.
Market quality	Liquidity, spreads, depth, concentration, manipulation risk, settlement venue quality, and source-market reliability.	Position limits, trading limits, surveillance escalation, or denial of listing.
Fork and replay	Chain-selection policy, replay protection, migration mechanics, bridge or wrapper exposure, and customer notice plan.	Trading may not begin without executable fork and replay controls.
Oracle and index dependency	Benchmark methodology, source venue ACP status, data signing controls, stale-price rules, and outlier handling.	Product, margin, or settlement use is denied where valuation cannot remain reliable during a cryptographic event.
Margin and collateral	Haircut methodology, concentration limits, liquidation test, custodian transfer test, and default-management treatment.	Asset may trade but remain ineligible for margin collateral, default-fund contribution, or delivery.
Disclosure	Product documents, risk factors, website disclosures, customer notices, and marketing controls.	Listing or product launch is delayed until disclosures match the ACP record.

11.3 Halt, Suspension, and Delisting Triggers

A market operator must maintain objective triggers for trading halts, creation and redemption suspensions, margin restrictions, clearing restrictions, index disruption events, and delisting reviews. The response must be proportional, but the operator may not wait for confirmed loss where a credible CRQC event, replay vector, custody failure, or ACP downgrade threatens settlement finality or orderly liquidation.

Trigger	Initial action	Escalating action
ACP downgrade or ACP-U finding	Public notice, admission review, and restriction on new PQC claims.	Halt, collateral ineligibility, product suspension, or delisting review.
Credible CRQC or signature-forgery event	Emergency risk meeting, withdrawal and settlement review, customer notice, and regulator notice.	Trading halt, liquidation controls, creation/redemption suspension, or market-wide incident procedure.
Legacy or nonconforming transfer path discovered	Suspend ACP-Q3 treatment and require negative vectors.	Delisting review or product restriction if the path can move production value.

Trigger	Initial action	Escalating action
Custodian transfer failure	Restrict new deposits, creations, collateral calls, or delivery obligations.	Transfer-out plan, custody replacement, product suspension, or wind-down.
Fork, replay, or migration instability	Activate chain-selection, deposit quarantine, replay testing, and disclosure controls.	Halt affected markets or suspend affected products until settlement finality is restored.
Oracle, index, or reference-price disruption	Invoke stale-price, outlier, fallback, and calculation-agent procedures.	Index disruption event, settlement-price adjustment, liquidation pause, or product suspension.
Margin liquidation failure	Increase margin, reduce collateral value, restrict new positions, and notify clearing members or customers.	Collateral ineligibility, default-management escalation, or position reduction order where authorized.
Material disclosure failure	Corrective disclosure and halt consideration.	Delisting review, enforcement referral, or disqualification from safe-harbor treatment.

11.4 Fork, Replay, Migration, and Wrapped-Asset Controls

Fork controls must be written before listing or clearing begins. The operator must identify who decides which chain is supported, how deposits and withdrawals are quarantined, how replay risk is tested, how open orders and derivatives are treated, how ETP creations and redemptions are handled, and how wrapped or bridged forms are mapped to the base asset.

For BTX-style assets, the central question is whether the active chain preserves production post-quantum transfer authority and rejects nonconforming paths. For account-based networks, bridges, app-chains, or wrapped assets, the same review covers validator keys, bridge administrator keys, smart-contract upgrade keys, sequencer keys, oracle keys, and wrapper redemption keys.

Event	Required controls	Market consequence
Planned PQC migration	Activation height, migration window, supported wallets, replay protection, customer notice, and transfer test.	Conditioned continued listing, temporary deposit controls, or staged product reopening.
Contentious fork	Chain-selection criteria, ACP comparison, liquidity review, custody support, and index treatment.	Halt or split treatment until the supported chain and product rights are disclosed.
Replay vulnerability	Cross-chain replay vectors, domain separation, transaction quarantine, and withdrawal testing.	Deposit and withdrawal suspension for affected chains until replay protection is proven.
Wrapped-asset dependency	Base-chain ACP, wrapper contract controls, bridge key ACP, redemption test, and issuer reserve proof.	Wrapper may receive a lower ACP treatment than the base asset if bridge or redemption keys remain vulnerable.
Emergency CRQC fork	Governance authority, customer priority, no-insider-preference rule, and post-event audit.	Halt, restrict, or delist if the fork changes economic rights without public, non-discretionary procedures.

11.5 Oracle, Index, and Reference-Price Controls

Oracle and index controls are market-integrity controls. A product can have strong custody while still failing if NAV, settlement, margin, or liquidation depends on a benchmark that sources prices from venues unable to process deposits, withdrawals, forks, or replay events during a cryptographic incident.

Control	Operational requirement	Failure response
Source venue ACP review	Classify source venues by supported asset ACP, custody capability, withdrawal status, and fork policy.	Exclude or de-weight venues that cannot support settlement during ACP-relevant events.
Oracle signing and admin keys	Inventory oracle, publisher, administrator, emergency, and upgrade keys in the CBOM.	Treat vulnerable oracle control keys as market infrastructure exposure even if the base asset is ACP-Q3.
Index methodology	Define source eligibility, outlier removal, stale data, fork treatment, airdrops, wrappers, and disruption events.	Trigger calculation-agent review and public methodology notice after material asset or venue events.
NAV and settlement prices	Tie ETP NAV, futures settlement, swaps valuation, and liquidation marks to documented benchmark controls.	Suspend creations, redemptions, settlement-price use, or liquidations where the price is not reliable.
Data integrity monitoring	Monitor price divergence, source outages, unexplained withdrawal suspensions, and chain-specific anomalies.	Escalate to halt, margin, and disclosure procedures when data quality indicates transfer-authority stress.

11.6 Margin, Collateral, Clearing, and Default Management

Margin and collateral treatment must follow ACP status. A DCO, FCM, clearing member, or product sponsor may not treat a digital asset as high-quality liquid collateral merely because it is actively traded. Transfer-authority weakness affects liquidation speed, settlement finality, custody control, wrong-way risk, and default-management reliability.

ACP	Collateral treatment	Margin controls	Clearing consequence
ACP-Q3	Eligible if custody, liquidation, and oracle controls pass.	Base haircut plus ordinary liquidity, concentration, and volatility add-ons.	May support cleared exposure if default-management transfers are tested.
ACP-Q2	Eligible only with hybrid-risk disclosure and downgrade monitoring.	Quantum-risk add-on, concentration cap, and annual or event-driven transfer test.	Clearing permitted only if downgrade or bypass triggers cause immediate eligibility review.
ACP-Q1	Restricted eligibility during transition period.	High add-on, low concentration cap, milestone monitoring, and liquidation drill.	Not acceptable for default fund use absent specific relief and documented customer-protection basis.
ACP-Q0	Generally ineligible for new collateral programs.	Legacy run-off only, heightened disclosure, and conservative liquidation assumptions.	No new clearing eligibility after phase-in dates absent relief.
ACP-U	Treated as ACP-Q0 until evidence is complete.	No collateral credit beyond approved wind-down treatment.	Clearing or margin application is incomplete.

11.7 Operator Variants

Different market operators apply the same ACP framework through different regulatory functions. The obligation is function-specific: an exchange focuses on listing and trading integrity, a DCO on clearing and default management, an FCM on customer margin and segregation, an ETP sponsor on disclosure and creations/redemptions, and a non-US operator on equivalence and cross-border incident coordination.

Operator	Specific obligation	Evidence record
Exchange or trading venue	Adopt listing, halt, delisting, surveillance, fork, replay, and disclosure standards tied to ACP status.	Listing file, ACP record, source-market review, custody transfer tests, and halt playbook.
DCM	Review underlier, deliverable, settlement index, forks, oracle dependencies, and manipulation susceptibility before certification or listing.	Product certification file, settlement methodology, ACP analysis, and disruption procedures.
SEF	Review swap references, execution protocols, oracle inputs, bilateral settlement assumptions, and emergency market procedures.	Product terms, benchmark map, counterparty notice procedure, and ACP-triggered trading controls.
DCO	Set collateral eligibility, margin add-ons, default-management transfer tests, settlement controls, and member notice rules.	Collateral policy, margin model change log, liquidation drill, custody test, and default playbook.
FCM	Disclose customer exposure, apply collateral haircuts, monitor customer asset segregation, and support withdrawal or transfer under stress.	Customer disclosure, margin file, segregation controls, custodian attestation, and transfer-out plan.
ETP sponsor or fund	Maintain underlier ACP review, custodian readiness, creation/redemption controls, NAV controls, fork treatment, and shareholder disclosure.	Registration disclosure, index methodology, custody test, authorized participant procedure, and incident notice template.
Index, oracle, or benchmark operator	Maintain source eligibility, data signing, key controls, stale-price rules, fork treatment, and disruption procedures.	Methodology, CBOM, source venue review, calculation logs, and administrator attestation.
Non-US venue, CCP, or product operator	Demonstrate comparable ACP controls, home-regulator reporting, local law compatibility, and cross-border incident communication.	Equivalence memo, local rule map, regulator contact plan, and U.S. market-impact analysis where U.S. persons or products are affected.

11.8 Market Operator Control Flow

Use this flow as the operator's control loop: admission review sets the evidence boundary, ongoing operation creates surveillance data, reportable events move into consequence review, and the resulting action feeds back into admission conditions and market operation.

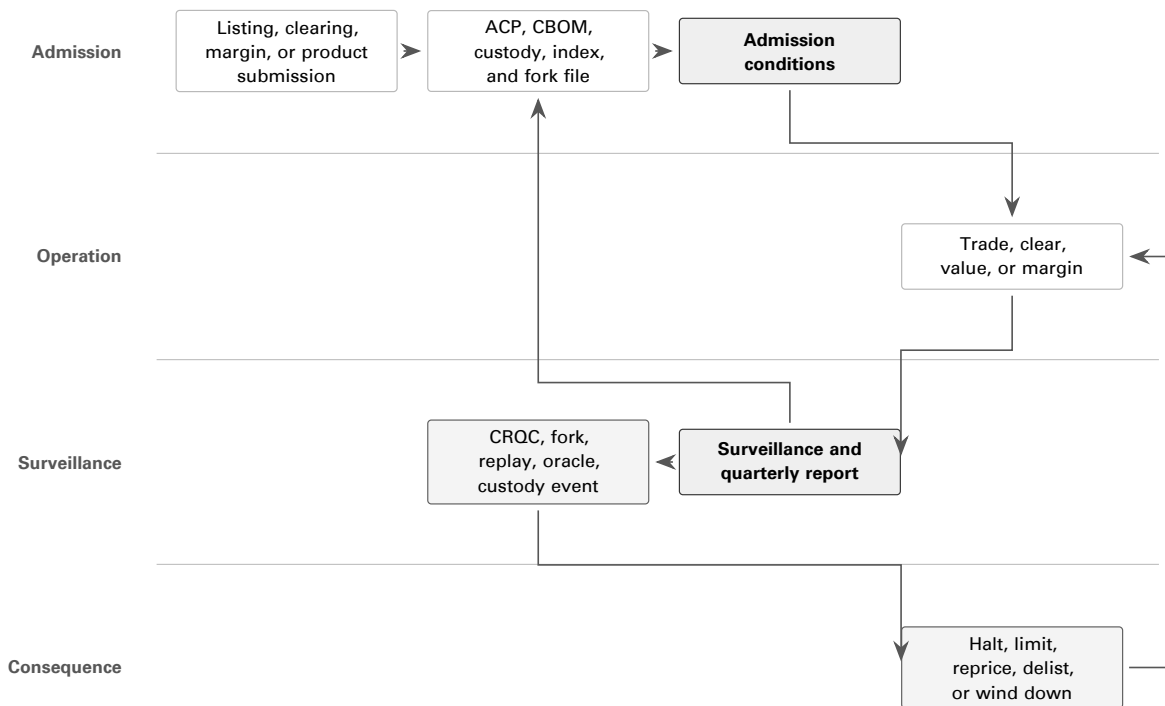


Figure 9: Market-operator control flow linking admission review, ongoing surveillance, quarterly reporting, and ACP-triggered market consequences.

12 Examination Consequences

The ACP framework is useful only if a classification changes product treatment. The following consequences turn the evidence record into supervisory action while preserving a path for good-faith migration. They are written as model outcomes a Commission, exchange, DCO, FCM, broker-dealer, adviser, custodian, or ETP sponsor can apply through existing disclosure, listing, custody, collateral, clearing, system-safeguards, and anti-fraud authorities.

Trigger	Immediate consequence	Escalating consequence
Unsupported quantum-safe claim	Corrective disclosure and cessation of the claim.	Enforcement referral for materially misleading statements or omissions.
Incomplete CBOM or evidence record	ACP-U treatment and suspension of post-quantum eligibility review.	Product application, custody onboarding, or collateral approval treated as incomplete.
ACP-Q0 asset in new sponsored product	Heightened risk disclosure and eligibility restriction.	New listings, collateral acceptance, or omnibus custody limited or denied after phase-in dates absent relief.
Missed ACP-Q1 milestone	Downgrade review and customer/product notice.	Downgrade to ACP-Q0 and restriction on new exposure.
Hybrid downgrade or bypass in ACP-Q2 asset	Immediate remediation plan and incident notice.	Suspension of ACP-Q2 treatment, trading halt consideration, collateral haircut increase, or withdrawal of eligibility.
Custodian cannot demonstrate transfer, sweep, recovery, or wind-down	Restriction on new custody exposure and required remediation test.	Customer transfer plan, withdrawal support, or custody relationship wind-down.
Material CRQC, fork, replay, bridge, validator, governance, or custody event	Incident report, disclosure update, and emergency risk review.	Trading halt, collateral ineligibility, liquidation plan, delisting review, or enforcement referral.
False officer attestation or concealed dependency	Corrective filing and supervisory escalation.	Enforcement referral, civil penalties, bar or suspension remedies where authorized, and disqualification from safe-harbor treatment.

13 Regulator Execution Toolkit

This section is the operating package. It converts the BTX-anchored ACP framework into checklists, scorecards, workpapers, and implementation controls that a regulator can use without creating a separate manual. The tables are designed to be copied into examination requests, listing-standard amendments, custody onboarding reviews, ETP comment letters, collateral eligibility procedures, and supervisory workpapers.

13.1 Program Charter Checklist

The charter checklist is the regulator’s internal launch document. It identifies the legal basis, covered persons, covered assets, evidence standard, and continuing-supervision commitments before staff begin collecting ACP submissions. Completing this checklist first prevents the program from becoming a series of ad hoc product comments or custody conditions with inconsistent terminology.

	Control	Execution requirement
☐	Legal basis	Identify the domestic authority used to require ACP evidence, including product approval, listing, disclosure, custody, collateral, clearing, market-infrastructure, prudential, cyber-resilience, anti-fraud, or consumer-protection authority.
☐	Covered persons	Define the regulated persons in scope: issuers, sponsors, exchanges, trading venues, clearing organizations, custodians, brokers, advisers, FCMs, DCOs, index providers, wallet providers, bridge operators, validators, or other regulated intermediaries.
☐	Covered assets	Identify digital assets, wrapped assets, bridges, staking or validator systems, collateral assets, ETP underliers, derivative underliers, deliverables, reserves, and custody assets subject to review.
☐	Reference benchmark	State that ACP-Q3 is measured by BTX’s reference functions or by functionally equivalent or superior production post-quantum transfer authority.
☐	Evidence standard	Require a claim boundary, CBOM, vector package, audit record, custody/wallet test report, network-operational evidence, disclosure matrix, and management attestation for every formal ACP claim.
☐	Product consequence	Tie ACP results to concrete actions: application completeness, listing eligibility, custody onboarding, collateral haircuts, disclosure updates, remediation milestones, trading halt triggers, and enforcement referrals.
☐	Continuing supervision	Require annual attestation, quarterly status reporting for non-Q3 assets, immediate material-change notice, and incident reporting after cryptographic, fork, bridge, custody, validator, or governance events.

13.2 Applicability Triage Checklist

Triage determines whether a product, venue, custodian, collateral program, or asset use enters the ACP review process. The checklist is intentionally framed as yes/no questions so front-line staff can route filings consistently before a full examination begins. A no answer does not create a permanent exclusion; it records the current basis for monitoring and should be revisited after a material change in asset use, marketing claims, custody support, or cryptographic dependency.

	Question	Yes result	No result
☐	Is the digital asset used in a regulated product, listing, custody program, collateral program, derivative, clearing process, index, trust, fund, or sponsored retail exposure?	ACP filing required before approval, onboarding, continued eligibility, or material amendment.	Monitor if no regulated use exists; prohibit unsupported PQC marketing by regulated persons.
☐	Does asset ownership, transfer, validation, governance, staking, bridge custody, oracle control, or recovery rely on public-key cryptography?	CBOM and transfer-authority mapping required.	Document exclusion and reassess if cryptographic control is added.
☐	Does the ordinary production transfer path rely on ECDSA, Schnorr over classical curves, EdDSA, RSA, DSA, ECDH, or another quantum-vulnerable asymmetric primitive?	Initial classification is ACP-Q0 unless transition or hybrid evidence supports Q1 or Q2.	Continue standards, vector, custody, and operational evidence review.
☐	Is a post-quantum or hybrid mechanism active in production?	Require production validation evidence, negative vectors, downgrade controls, and custody/wallet tests.	Treat as transition-only; prohibit ACP-Q3 statements.
☐	Is the asset marketed as quantum-safe, post-quantum, quantum-resistant, NIST-approved, FIPS-based, or equivalent?	Require evidence-level review and corrective disclosure for unsupported claims.	Require risk disclosure if the product is otherwise in scope.
☐	Could failure affect retail customers, fund shareholders, margin, liquidation, settlement, custody, market integrity, or default management?	Apply heightened supervisory review and emergency playbook requirements.	Maintain baseline ACP documentation and annual reassessment.

13.3 Critical Failure Gates

The numerical scorecard below never overrides a critical failure. A project that triggers any critical gate is not ACP-Q3 until the defect is remediated, retested, and supported by updated evidence.

Gate	Critical failure	Required regulator action
G1	Production ordinary transfer can be completed through an unmitigated quantum-vulnerable signature path.	Deny ACP-Q3; classify as ACP-Q0, Q1, or Q2 only if supported by transition or hybrid evidence.
G2	The claimed post-quantum mechanism is wallet-only, policy-only, testnet-only, optional, or bypassable in production.	Deny production conformance; require production validation proof and negative consensus vectors.
G3	Claim boundary, release, binary, activation state, or deployed network cannot be uniquely identified.	Treat application as incomplete; require corrected boundary and reproducible artifact record.

Gate	Critical failure	Required regulator action
G4	CBOM omits a material signing, custody, bridge, governance, validator, oracle, HSM, MPC, wallet, API, or recovery dependency.	Treat evidence package as incomplete; require updated CBOM and dependency attestations.
G5	Negative vectors accept a legacy signature, wrong commitment, malformed key, replay, downgrade, or wrong activation state.	Deny certification; require remediation, retest, and independent vector attestation.
G6	The filer implies NIST, FIPS, regulator, or laboratory approval beyond the evidence obtained.	Require corrective disclosure and marketing controls; consider enforcement referral for material misstatement.
G7	Custodian cannot demonstrate withdrawal, sweep, recovery, wind-down, or customer transfer under the claimed model.	Restrict custody onboarding or continued custody treatment until transfer testing passes.
G8	A material CRQC, cryptographic break, fork, bridge, validator, governance, or custody incident lacks an executable emergency response plan.	Require incident plan, customer/product notice protocol, and emergency eligibility review.

13.4 ACP Scorecard

The scorecard gives examiners a repeatable way to compare projects. Scores are assigned only after critical gates are cleared. BTX supplies the ACP-Q3 reference profile: a project receives full credit in a domain only when its evidence is comparable to or stronger than BTX's reference functions for that domain.

ID	Domain	Pts	Scoring rule
S1	Claim boundary	5	Full credit for exact asset, network, release, commit, binary, activation rule, product use, custody system, and exclusions. Partial credit for minor omissions. Zero for ambiguous scope.
S2	CBOM completeness	10	Full credit for protocol, wallet, custody, vendor, HSM, MPC, API, bridge, governance, validator, transport, build, recovery, and residual-exposure inventory.
S3	Transfer authority	15	Full credit when ordinary production transfer does not require a quantum-vulnerable signature and is benchmarked to controls functionally equivalent or superior to BTX's reference functions. Hybrid-active designs may receive partial credit.
S4	Production enforcement	12	Full credit for consensus, contract, bridge, or validator rules that enforce the claimed transfer model in production and reject nonconforming transfer paths.
S5	Standards and claim evidence	10	Full credit when FIPS 203/204/205 or successor mappings, parameter sets, ACVP/CAVP evidence where claimed, and FIPS 140-3 boundaries where claimed are precise.
S6	Test vectors and negative tests	10	Full credit for signed manifests covering KATs, protocol vectors, invalid signatures, malformed inputs, wrong commitments, replay, downgrade, activation, and interoperability cases.
S7	Wallet and custody operability	10	Full credit for demonstrated deposit, withdrawal, sweep, recovery, wind-down, hardware-wallet, HSM/MPC, API, and customer-notice workflows.
S8	Downgrade, replay, and migration controls	8	Full credit for fork, replay, downgrade, migration, bridge, wrapped-asset, and emergency CRQC controls that preserve customer and market integrity.

ID	Domain	Pts	Scoring rule
S9	Network and market operations	10	Full credit for activation status, node/miner/validator adoption, liquidity, fee/size impact, storage/verification cost, orphan/re-org/finality evidence, and exchange/custodian readiness.
S10	Disclosure and attestation	10	Full credit for officer certification, public risk factors, ACP status, limitation statements, marketing controls, material-change reporting, and regulator-facing determination language.
		Total 100	Score cannot override critical failure gates.

Score band	ACP result	Conditions
90–100	ACP-Q3	No critical gates triggered; production ordinary transfer authority is post-quantum and functionally equivalent or superior to BTX's reference functions; all mandatory vectors and attestations pass.
75–89	ACP-Q2	No critical gates inconsistent with hybrid status; production transfer authority is hybrid-active and downgrade-resistant; residual classical dependency is disclosed.
50–74	ACP-Q1	Transition plan is credible, public, dated, and testable; production transfer remains quantum-vulnerable; marketing is limited to transition status.
0–49	ACP-Q0	Evidence is incomplete or production transfer authority remains quantum-vulnerable without credible transition controls.
Any score	ACP-U	Claim boundary or required evidence is too incomplete to examine. ACP-U is treated as ACP-Q0 for eligibility until cured.

13.5 Examiner Workpaper

The workpaper is the staff record for a specific examination or filing review. It should be completed after triage, critical-gate review, and evidence intake, and it should be retained with the claim boundary, scorecard, vector records, attestation materials, and any conditions imposed. The table is not merely a checklist; it is the audit trail showing how the examiner moved from evidence to ACP classification.

	Workpaper item	Evidence source	Result
☐	Confirm legal entity, responsible officer, asset, product, and claim boundary.	Filing cover letter, management attestation, repository/release record.	Complete / deficient / not applicable.
☐	Reconcile repository, commit, release tag, binary hash, deployment, and activation state.	Source control, signed releases, node data, chain state, build logs.	Complete / deficient / not applicable.
☐	Review transfer-authority map against BTX's reference functions.	Protocol specification, code exhibits, consensus or contract rules.	Complete / deficient / not applicable.
☐	Test critical negative vectors.	Vector manifest, lab logs, examiner rerun, rejection codes.	Pass / fail / retest required.
☐	Review wallet and custody operability.	Wallet test report, API samples, HSM/MPC evidence, sweep/recovery logs.	Pass / fail / retest required.
☐	Review public-key exposure and residual legacy exposure.	Chain analytics, custody inventory, bridge lock records, validator/governance key inventory.	Complete / deficient / not applicable.
☐	Review market and operational readiness.	Liquidity data, node/miner/validator adoption, exchange/custody readiness, incident drills.	Complete / deficient / not applicable.
☐	Review disclosure, marketing, and limitation language.	Prospectus, risk factors, website, sales material, listing notices, customer notices.	Accurate / corrective disclosure required.
☐	Assign score, ACP class, conditions, and next review date.	Scorecard, critical gates, examiner memo, determination order.	ACP-Q0 / Q1 / Q2 / Q3 / U.

13.6 Asset-Type Review Modules

Different asset designs expose different transfer-authority surfaces, so the common ACP framework must be applied through asset-type modules. This table tells examiners what additional surfaces to inspect after the general gates have been applied. The modules are additive: a wrapped asset used in an ETP, for example, may require bridge, custody, product, and market-infrastructure tests at the same time.

Asset or activity	Additional review surface	Minimum execution tests
Bitcoin-style output-accounting asset	Address types, public-key exposure, wallet defaults, signing paths, migration mechanics, and consensus validation.	Valid PQ transfer, invalid legacy output, invalid legacy signature, wrong commitment, replay /-fork boundary, wallet sweep, and recovery transfer.
Account-based network	Account signatures, validator keys, staking withdrawal keys, smart-contract admin keys, bridges, L2s, emergency multisigs, and governance execution keys.	Account transfer, validator rotation, staking withdrawal, contract-admin action, bridge withdrawal, replay / downgrade rejection, and emergency governance drill.
Bridge or wrapped asset	Lockbox keys, mint/burn controls, relay/oracle signatures, reserve attestations, replay protection, chain identifier binding, and emergency halt authority.	Lock, mint, burn, withdrawal, proof replay rejection, wrong-chain rejection, reserve reconciliation, and operator key rotation.
Custodian or wallet	Key generation, HSM/MPC policy, signing, withdrawal, sweep, recovery, wind-down, firmware/software signing, vendor dependencies, and customer notice.	Deposit, withdrawal, batch sweep, cold recovery, lost device workflow, wind-down transfer, incident freeze/release, and customer notification exercise.
ETP, fund, or trust	Underlying asset ACP, custodian ACP, creations/redemptions, NAV, forks, airdrops, liquidity, disclosure, and shareholder communication.	Creation/redemption stress, custodian transfer test, fork/replay tabletop, NAV impact analysis, risk-factor update, and emergency disclosure drill.
Derivative or collateral program	Underlier ACP, deliverability, cash-settlement index, collateral transfer, haircuts, liquidation, default management, and concentration risk.	Collateral deposit/withdrawal, liquidation simulation, haircut recalculation, default auction playbook, index disruption test, and margin call stress.

13.7 Quarterly Reporting Dashboard

Quarterly reporting keeps an ACP classification current after the initial determination. The dashboard is designed for assets, custodians, and market operators that remain under conditions, maintain transition milestones, or support regulated products with continuing public exposure. It should be used to detect drift: stale vectors, missed milestones, growing residual legacy exposure, declining wallet or custody support, deteriorating liquidity, or an outdated CRQC incident plan.

Metric	Required content	Threshold action
ACP status	Current ACP, score, claim boundary, release, activation height, and evidence date.	Downgrade review after material change or missed milestone.
Residual legacy exposure	Value, supply percentage, addresses/accounts, validators, governance keys, bridge keys, and custody balances still controlled by quantum-vulnerable keys.	Heightened disclosure and mitigation plan if exposure increases or remains above filed milestones.
Vector and audit status	Last vector run, failing cases, remediation commits, lab attestations, and open audit findings by severity.	Certification suspension for critical open findings or failed mandatory vectors.
Wallet and custody readiness	Supported wallets, HSM/MPC systems, hardware signers, exchange/custody integrations, withdrawal tests, recovery tests, and wind-down tests.	Custody restriction if production transfer or recovery cannot be demonstrated.
Market operation	Liquidity, spreads, volume, fee impact, block/state growth, verification cost, reorg/finality data, and exchange readiness.	Product limit, haircut review, or trading-halt analysis after material deterioration.
Incident readiness	CRQC triggers, tabletop date, responsible officers, contact tree, customer notices, emergency release process, and regulator notice path.	Corrective action if tabletop is stale, incomplete, or failed.

13.8 Jurisdiction-Neutral Adoption Map

Regulators outside the United States can adopt the same execution model by replacing SEC/CFTC-specific terms with domestic authority labels. The supervisory core does not change: the regulator still fixes the claim boundary, requires a cryptographic inventory and test record, scores transfer-authority conformance against BTX's reference functions, and ties the ACP result to product, custody, market-infrastructure, disclosure, or enforcement consequences.

Regulatory function	Domestic implementation hook	ACP execution action
Product authorization	Fund, ETP, exchange-traded product, structured product, public-offer, prospectus, listing, or admission-to-trading authority.	Require ACP, CBOM, disclosure matrix, custody transfer test, and CRQC playbook before approval or continued listing.
Market infrastructure	Trading venue, clearing, settlement, depository, payment, index, benchmark, or system-resilience authority.	Require transfer-authority risk review, settlement/fork playbook, index-disruption controls, and incident reporting.
Custody and safeguarding	Custodian, wallet provider, broker, trust company, depository, client-asset, prudential, outsourcing, or operational-resilience authority.	Require custody ACP, wallet/HSM/MPC CBOM, withdrawal/sweep/recovery tests, vendor attestations, and customer notice procedures.
Collateral and margin	Margin, eligible collateral, haircut, liquidation, default-management, capital, or risk-management authority.	Classify collateral assets by ACP, adjust haircuts, test transfer/liquidation, and impose concentration limits for ACP-Q0/Q1 assets.

Regulatory function	Domestic implementation hook	ACP execution action
Disclosure and conduct	Prospectus, risk-factor, fair dealing, suitability, marketing, anti-fraud, consumer protection, or client communication authority.	Prohibit unsupported PQC claims, require ACP status, disclose residual exposure, and mandate corrective notices after material events.
Cyber and operational resilience	Technology-risk, incident-reporting, outsourcing, third-party-risk, business-continuity, or critical-infrastructure authority.	Require CBOM, vector reruns, dependency attestations, emergency CRQC drills, and material-change notices.
Cross-border supervision	Home-host coordination, substituted compliance, passporting, equivalence, recognition, or memorandum of understanding.	Share ACP determinations, claim boundaries, incident notices, evidence hashes, and conditions without requiring identical domestic legal categories.

13.9 Regulator Decision Record

The decision record is the final examination artifact. It should be completed only after the examiner has fixed the claim boundary, verified the evidence package, applied the critical gates, and assigned an ACP score. The record gives the regulated person a durable classification, conditions, next-review date, and material-change notice obligation without implying any endorsement of investment value.

EXAMPLE OUTPUT		SPECIMEN RECORD
ACP Determination Record		
Authority	[domestic legal authority]	
Regulated person	[entity]	
Asset or product	[asset/product]	
Claim boundary	[network, release, commit, binary, activation state, custody systems, products, exclusions]	
Evidence reviewed	[CBOM, vector manifests, audit reports, custody tests, operational evidence, disclosures, attestations]	
Critical gates	[passed/failed]	
Score	[0–100]	
ACP classification	[ACP-U/Q0/Q1/Q2/Q3]	
Conditions	[milestones, disclosure, testing, reporting, eligibility limits]	
Next review date	[date]	
Material-change notice required	[yes/no]	

Specimen 13: Final regulator decision record used to communicate ACP classification, conditions, review cadence, and material-change notice requirements.

14 Templates for Regulatory Adoption

14.1 ETP Disclosure Insert

Use this insert when a product sponsor, exchange, or reviewing staff needs risk-factor language that reflects the assigned ACP status and avoids unsupported claims of regulator approval. The insert should be updated whenever the claim boundary, custodian, underlier ACP status, fork treatment, or transition plan materially changes.

EXAMPLE OUTPUT		SPECIMEN RECORD
Crypto ETP Quantum Risk Disclosure		
Risk factor	The underlying digital asset and related network rely on cryptographic systems to authorize transfers, establish control of assets, secure network communications, and support custody.	
Quantum event	Developments in mathematics or computing, including a cryptographically relevant quantum computer, could reduce or eliminate the security assumptions underlying one or more of those systems.	
Potential impact	If a malicious actor could derive private keys from exposed public keys, forge signatures, compromise validator or governance keys, or exploit a migration weakness, the actor could misappropriate assets, impair settlement finality, disrupt the network, cause forks or replay events, affect liquidity, interfere with creation or redemption activity, and adversely affect the value of the Shares.	
ACP status	The Trust has classified the asset as [ACP-Q0/Q1/Q2/Q3] under its post-quantum conformance analysis.	
Limitation	This classification is not an approval by the SEC, CFTC, any exchange, or any other regulator.	

Specimen 14: ETP disclosure insert tying quantum-risk disclosure to the asset's ACP status and claim boundary.

14.2 CFTC Collateral Insert

Use this insert when a DCO, FCM, clearing member, or product sponsor proposes to accept a digital asset as margin, collateral, default-management resource, or liquidation asset. The provision makes ACP status part of collateral eligibility and gives the risk committee an event-driven reassessment trigger.

EXAMPLE OUTPUT		SPECIMEN RECORD
Collateral Eligibility Condition		
Before acceptance	A registered entity or registrant must document the asset's ACP, protocol-level transfer signature algorithms, custody transfer controls, exposed-public-key risk, liquidity and haircut impact, default-management procedures, and CRQC emergency triggers.	
Reassessment cadence	The entity must reassess the haircut and eligibility of the asset at least monthly.	
Event trigger	Immediate reassessment is required after a material cryptographic, custody, network, fork, validator, bridge, or market event.	

Specimen 15: Collateral eligibility condition tying margin and clearing treatment to ACP status, transfer controls, and event-driven reassessment.

14.3 Custody Attestation Insert

Use this insert when a custodian, wallet provider, ETP sponsor, broker-dealer, FCM, adviser, or market operator needs an officer-level custody representation. It is not a substitute for the custody scorecard; it is the management certification that the required protocol, wallet, HSM, MPC, vendor, recovery, and wind-down evidence exists inside the stated boundary.

EXAMPLE OUTPUT		SPECIMEN RECORD
Custodian Attestation		
Risk assessment	The custodian maintains a written PQC risk assessment for each covered digital asset.	
Dependencies	The custodian has identified residual quantum-vulnerable dependencies in protocol, wallet, custody, HSM, MPC, vendor, and recovery workflows.	
Operational tests	The custodian has tested material withdrawal, sweep, recovery, and wind-down processes.	
Disclosure	The custodian has disclosed material limitations to covered product sponsors and regulated counterparties.	
Authorized signatory	[name / title / date]	

Specimen 16: Custody attestation insert used as officer certification for custody, recovery, vendor, and disclosure controls.

15 Implementation Roadmap

The roadmap sequences adoption so a regulator can begin with disclosure and inventory controls, then move to pilots, product gates, custody and market consequences, and continuing surveillance. Domestic agencies can adopt the phases through rulemaking, staff guidance, exchange rules, supervisory letters, or product-specific conditions; non-US regulators can map the same phases onto their own licensing, recognition, custody, market-abuse, outsourcing, and operational-resilience authorities.

Phase	Regulatory action	Deliverables
0 to 90 days	Issue staff statement, examination risk alert, or joint request for information.	Stop unsupported quantum-safe marketing claims; require voluntary ACP submissions for new products; define CBOM expectations.
90 to 180 days	Adopt or propose interpretive guidance for ETPs, custody, collateral, and registered entities.	PQC migration lead, CBOM, asset inventory, initial ACP, disclosure update, CRQC playbook.
1 year	Add ACP to new product applications, listing standards, custody onboarding, DCO collateral policies, and FCM notices.	Product eligibility packet: ACP, audit plan, vectors, custody migration test, network evidence.
2027	Require production-like pilots.	Wallet/custody transfer pilots, fork/replay tabletop, customer notification and withdrawal tests.
2030	Align high-risk financial infrastructure with federal key-establishment expectations.	PQC or hybrid key establishment for APIs, custody operations, admin access, market connectivity, and settlement infrastructure.
2031	Align high-risk digital signature workflows.	PQC or hybrid signatures for custody approvals, firmware/software signing, governance, transfer authority where controlled by the covered person.
2035 backstop	Restrict ongoing ACP-Q0 exposure absent relief; accelerate before 2035 if CRQC, signature-forgery, AI-assisted quantum acceleration, regulator-guidance, or missed-milestone triggers arise.	Product limitations, delisting/suspension triggers, liquidation or conversion plans, final migration reports.

16 Conclusion

The federal government has recognized that post-quantum migration is a dated operational imperative. Financial-market digital-asset regulation has not yet applied that imperative to the asset-transfer layer. The gap is material because digital assets are not merely secured by cryptography; they are transferred, settled, valued, and custody-controlled by cryptography.

This framework establishes a BTX-anchored Asset Conformance Profile system backed by CBOMs, audit evidence, test vectors, custody controls, market-operational evidence, examiner checklists, scorecards, workpapers, international implementation mapping, examination determinations, and product-eligibility consequences. Bitcoin, Ethereum, Solana, HYPE/Hyperliquid, and other non-PQ assets are not banned by assertion; they are required to disclose and transition toward transfer-authority conformance functionally equivalent to BTX's reference functions or face product-eligibility consequences as federal PQC timelines mature.

BTX is the practical reference asset because it shows the path regulators need: preserve commodity-like properties while replacing vulnerable transfer authority with production-enforced post-quantum authorization and NIST-standardized post-quantum signature families. A regulator that anchors the framework in BTX has a tangible, live, Bitcoin-equivalent reference profile for what post-quantum digital commodity conformance should look like.

Source Base

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- SEC order approving generic listing standards for commodity-based trust shares, 90 FR 45414, Sept. 22, 2025
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- Korea MSIT, post-quantum cryptography transition master plan
- Korea FSC, financial-sector quantum security task force
- KISA, quantum-readiness statement
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- HKMA quantum preparedness survey
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- MAS advisory on cybersecurity risks associated with quantum computing
- MAS and Banque de France PQC experiment
- CSA guidance to organisations to be quantum ready
- ASD/ACSC Information Security Manual cryptography guidelines
- ASIC roadmap for digital-assets law reform implementation
- BTX clean source review used for code evidence. Commit f3c9eb7, tag v0.32.12; source evidence retained in the report workpapers.